Student Enrollment & Course Management System (SECMS)

Project work submitted to Thiruvalluvar University, Serkkadu, Vellore.

In partial fulfilment for the award of degree

BACHELOR OF SCIENCE

IN

Computer science

By

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Department of computer science

SHANMUGA INDUSTRIES ARTS &SCIENCE COLLEGE

Accredited with "B+" Grade by NAAC,

Certified Under Section 2(f) & 12B of the UGC Act 1956,

An ISO 9001:2015 Certified Institution

Permanently Affiliated to Thiruvalluvar University, Vellore,

Approved by Govt. of Tamil Nadu & AICTE,

Tiruvannamalai – 606603.

Student Enrollment & Course Management System(SECMS)…

Backlog item:

The Student Enrollment & Course Management System is a comprehensive Salesforce application designed to streamline and automate the academic enrollment operations for educational institute. This system enables administrators, enrollment officers, and instructors to efficiently manage student records, course offerings, enrollments, instructor assignments, approval processes,payments,andreporting—all with in a unified Salesforce environment.

1. INTRODUCTION

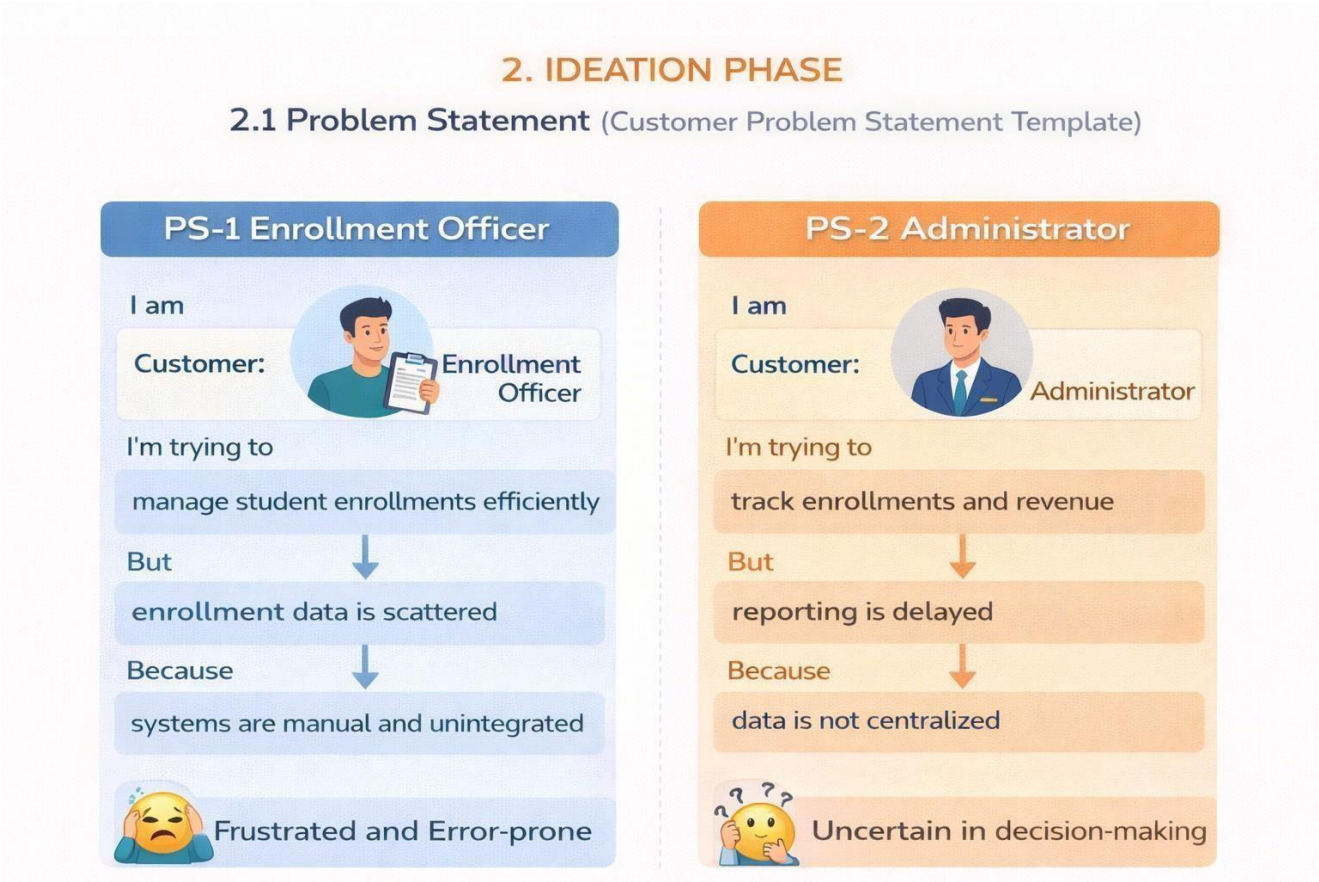
This project aims to centralize all student, course, and enrollment-related data within a single Salesforce application to ensure easy data access,structured workflows,and accurate reporting. The system allows Enrollment Officers to manage student admissions, course registrations,and communication effectively,while instructors and administrators gainfull transparency into course allocations and academic progress. The solution also enables automated instructor assignment,task creation,validation checks,and multi-level approval processes for enrollment management.

Purpose:

1. Accurate student data management
2. Structured enrollment workflows
3. Automated approvals and
4. validations
5. Transparentreportingandmonitoring

IDEATION PHASE

Problem Statement(Customer Problem Statement Template)

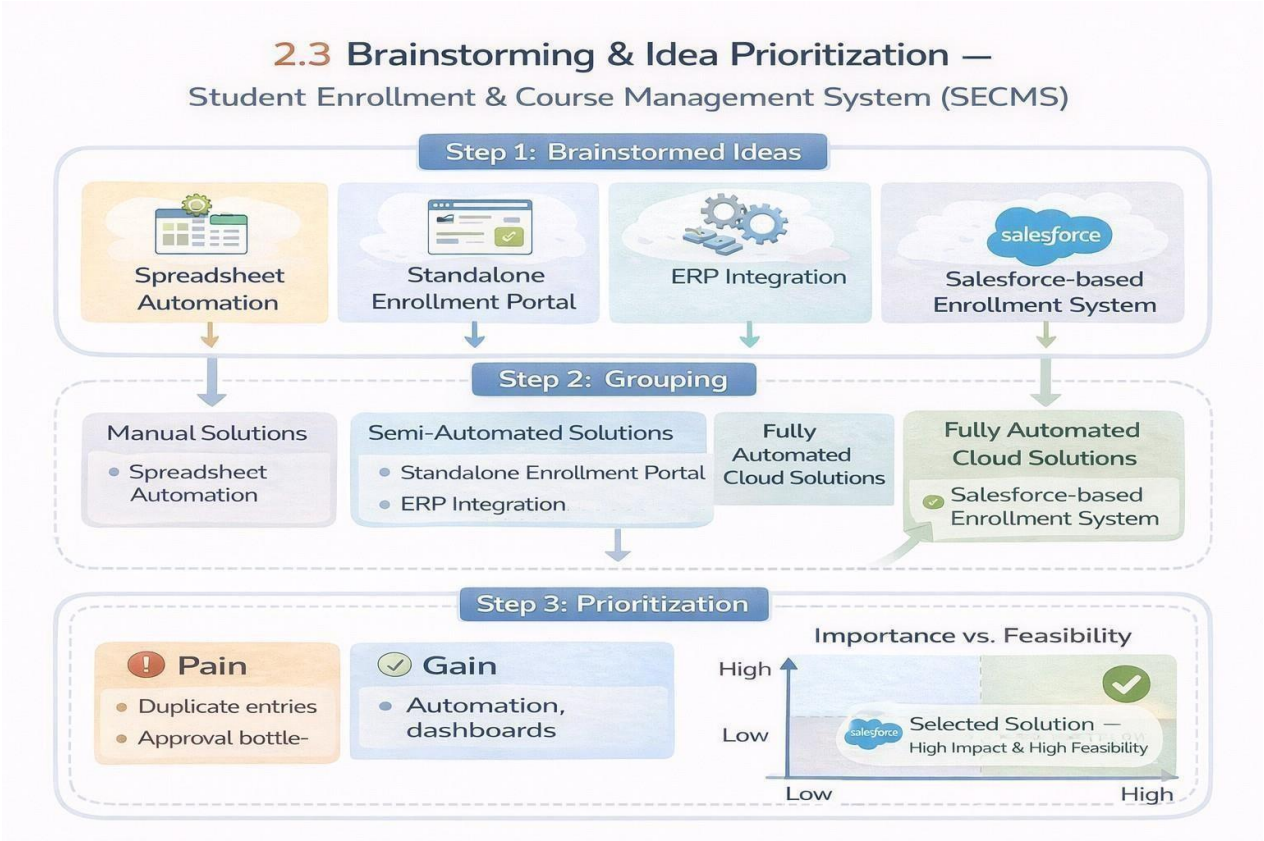


Empathy Map Canvas

UserPersona: Enrollment Officer

- **Think&Feel :**Concerned about double bookings and revenue loss
- **Hear :**Complaints from students and management
- **See :**Multiple spreadsheets andemails
- **Say&Do :**Rechecks data,follows upmanually
- **Pain :**Duplicate entries,approval bottle necks
- **Gain :**Automation,dashboards,fasterprocessing

Brainstorming & Idea Prioritization



2. REQUIREMENT ANALYSIS PHASE

- Awareness & Registration
- Course Selection
- Enrollment Request
- Fee Payment
- Approval
- Instructor Assignment
- Reporting & Completion

Solution Requirements

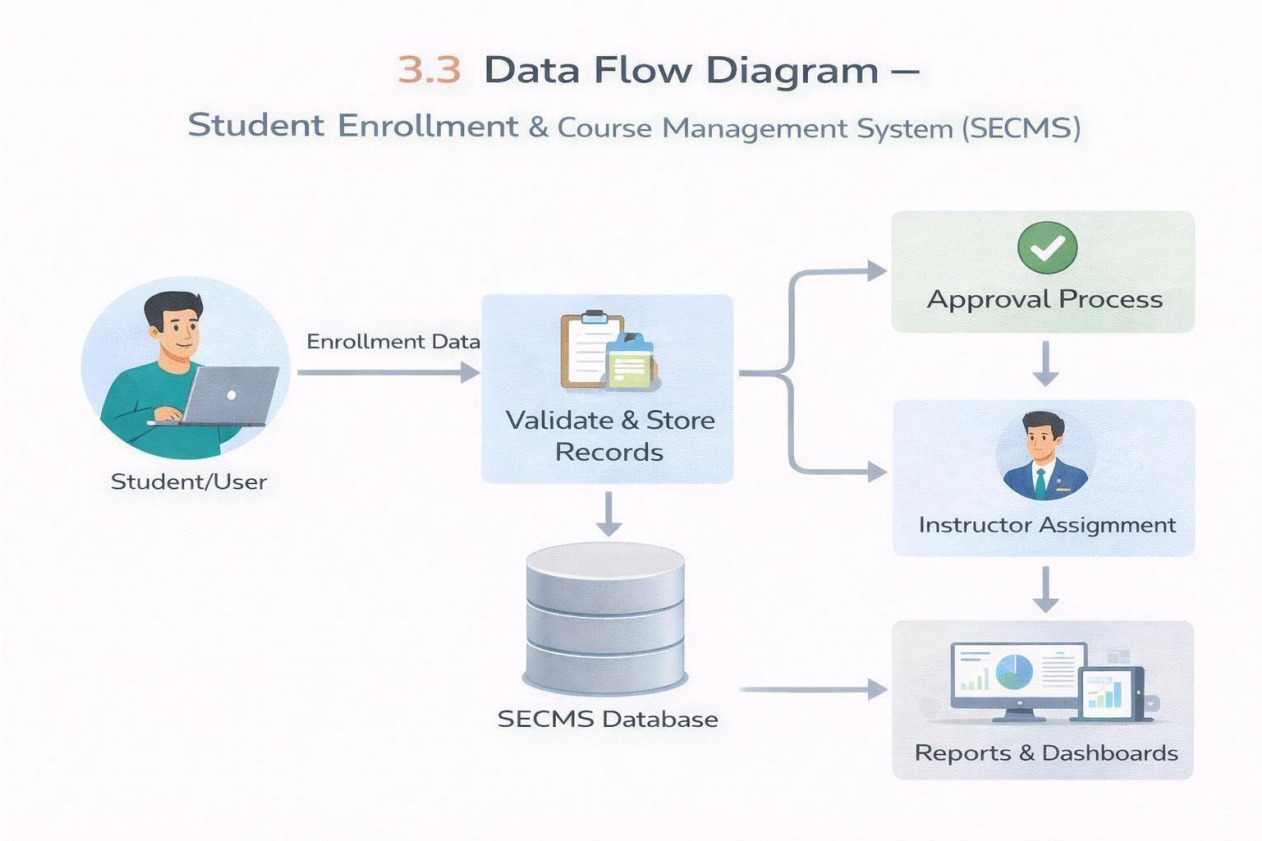
Functional Requirements

FRNo	Epic	Description
FR-1	Student Management	Manage student records
FR-2	Course Management	Create and manage courses
FR-3	Enrollment Processing	Request,approve,reject enrollments
FR-4	Instructor Assignment	Auto-assign instructors
FR-5	Reporting	Dashboards and reports

Non-Functional Requirements

NFR-1:Usability
NFR-2:Security
NFR-3:Reliability
NFR-4:Performance
NFR-5:Scalability
NFR-6:Availability

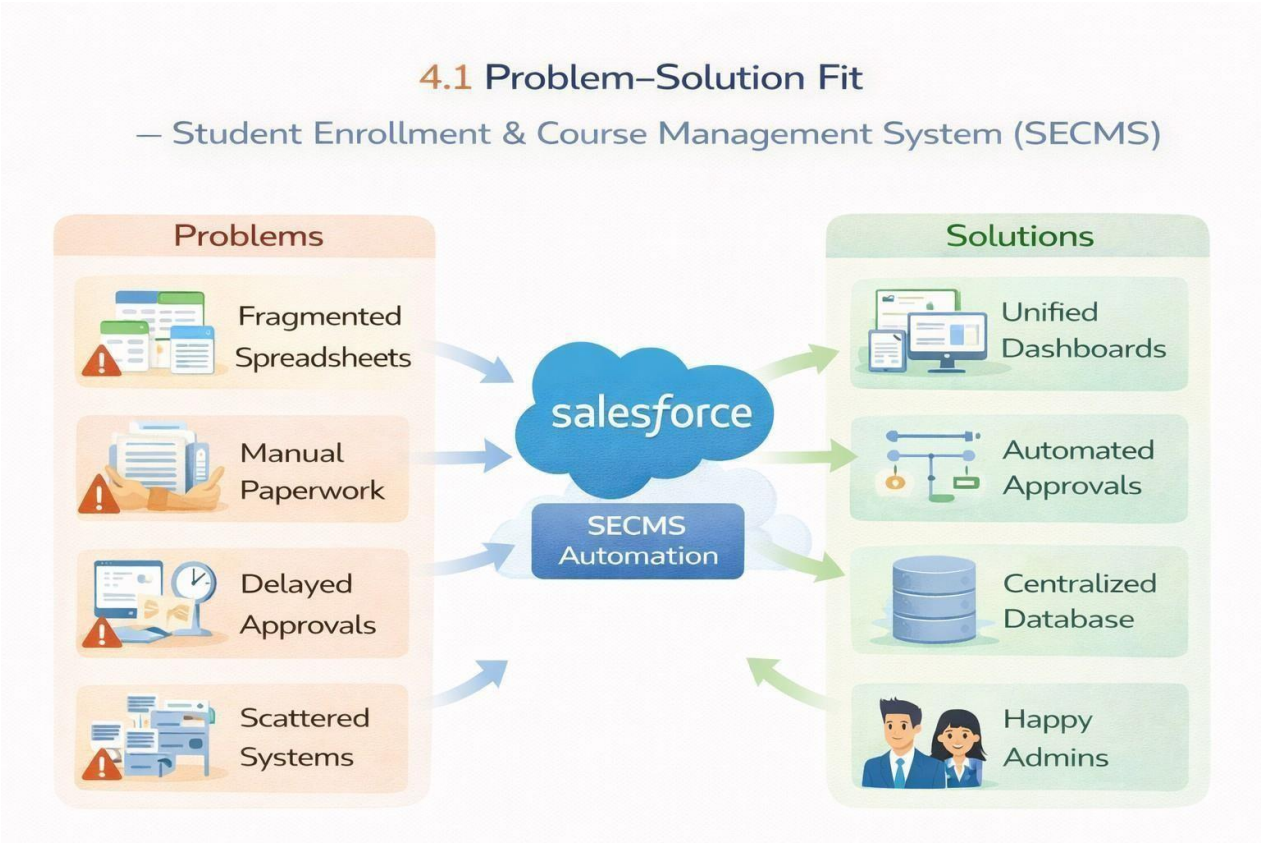
Data FlowDiagram



TechnologyStack

Layer	**Technology**
UI	SalesforceLightning
Logic	SalesforceFlows
Database	SalesforceObjects
Automation	Validation Rules,Approval Process
Reporting	Reports &Dashboards

PROJECT DESIGN PHASE



Proposed Solution

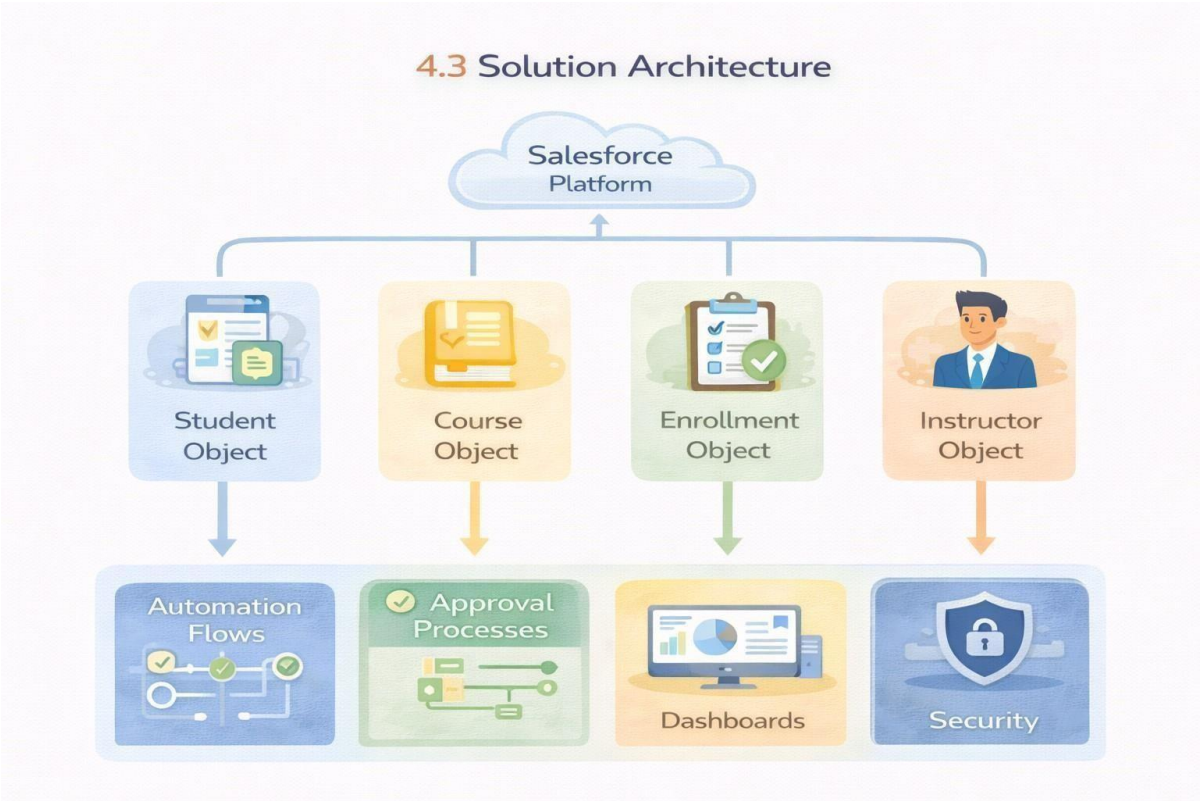
Functional Requirements

Parameter	**Description**
Problem	Manual enrollment handling
Solution	Salesforce-basedSECMS

Novelty	End-to-end automation
Social Impact	Improved student experience
Revenue Model	Institutional deployment
Scalability	Multi-department support

Solution Architecture

The architecture integrates Salesforce objects,automation flows,approval processes, dashboards, and security layers.



Project Planning

Sprint	Functional Requirement (Epic)	User Story Number	UserStory/Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create a Salesforce developer accountsothatIcanconfigureand deploy the Student Enrollment system..	1	High	J.ANGEL BHARATHI
Sprint	**Functional Requirement (Epic)**	**User Story Number**	**UserStory/Task**	**Story Points**	**Priority**	**Team Members**
			Course,Enrollment,andInstructor so that academic data can be storedandmanaged efficiently.			
Sprint-2	Data Modeling	USN-3	Asauser,Iwantdedicatedtabsfor each object so that I can easily navigate between students, courses, enrollments, and instructors.	3	High	D.GAYATHRI
Sprint-2	Data Modeling	USN-4	As an admin, I want to create relevant fields such as student details, course information, and enrollmentstatussothatcomplete records can be captured accurately.	5	High	M.KARTHIKA
Sprint-2	Data Modeling	USN-5	Asan enrollment officer, I want a centralizedLightningAppsothatI can access all academic operations from one interface.	3	High	M.KARTHIKA
Sprint-2	Data Modeling	USN-6	As an admin, I want to define relationships between students, courses,andenrollmentsso that thesystem canlogicallylinkand track academic activities.	5	Medium	J.ANGEL BHARATHI
Sprint-2	Data Modeling	USN-7	Asauser,Iwantcustomizedpage layouts so that only relevant informationisdisplayed,improving clarity and ease of data entry.	5	High	A.ABINAYA
Sprint-3	Automation	USN-8	As a system, I want to validate input data such as mandatory fieldsand enrollment limitsso that incorrectorincompleterecordsare prevented.	3	High	KARTHIKA.M
Sprint-3	Automation	USN-9	As a system, I want automated flows to handle enrollment processingandnotificationssothat manual effort is reduced and processesbecomefaster.	3	High	RAMYA.K

Sprint-3	Automation	USN-10	As an administrator, I want an approval process for enrollment requestssothatregistrationsare verified before confirmation.	5	High	ANGEL BHARATHI.j
Sprint-4	Security	USN-11	Asanadmin,Iwanttoconfigure role-basedsecuritysothatusers can only access data relevant to their responsibilities.	5	High	ABINAYA.A
Sprint-5	Reports	USN-12	As a manager, I want reports on student enrollments, course registrations, and instructor assignmentssothatIcanmonitor academicoperations.	4	High	RAMYA.K
Sprint-6	Dashboards	USN-13	As a manager, I want visual dashboardsshowingenrollment	4	Medium	RAMYA.K

ProjectTracker,Velocity&BurndownChart:(4Marks)

Sprint	TotalStory Points	Duratio n	SprintStart Date	SprintEndDate (Planned)	StoryPoints Completed (as on PlannedEnd Date)	SprintRelease Date (Actual)
Sprint-1	20	6Days	24 Oct2022	29 Oct2022	20	29 Oct 2022
Sprint-2	20	6Days	31 Oct2022	05Nov2022	21	5nov 2022
Sprint-3	20	6Days	07Nov2022	12Nov2022	23	12Nov2022
Sprint-4	20	6Days	14Nov2022	19Nov2022	23	19Nov2022

Project Development Phase

Project Flow:

Milestone1 :SalesforceDeveloperAccountCreation
Milestone2 :CustomObjectCreation(Student, Course,Enrollment,Instructor)
Milestone 3 : Tabs Creation
Milestone4 :LightningAppCreation
Milestone 5 : Fields Creation
Milestone 6 : Create Relationships
Milestone7 :PageLayoutCustomization
Milestone 8 : Validation Rules
Milestone9 :Flows(Automation)
Milestone 10 : Approval Process
Milestone 11 : Reports
Milestone 12 : Dashboards
Milestone13 :SecuritySetup
Milestone 14 : Conclusion

Milestone1-Salesforce:

Introduction:

Are you new to Salesforce? Not sure exactly what it is, or how to use it? Don't know where you should start on your learning journey? If you've answered yes to any of these questions,then you're in the right place.This module is for you.
Welcome to Salesforce! Salesforce is game-changing technology, with a host of productivity-boosting features,that will help you sell smarter and faster.As you work toward your badge forth is module, we'll take you through these features and answer the question, "What is Salesforce, anyway?".

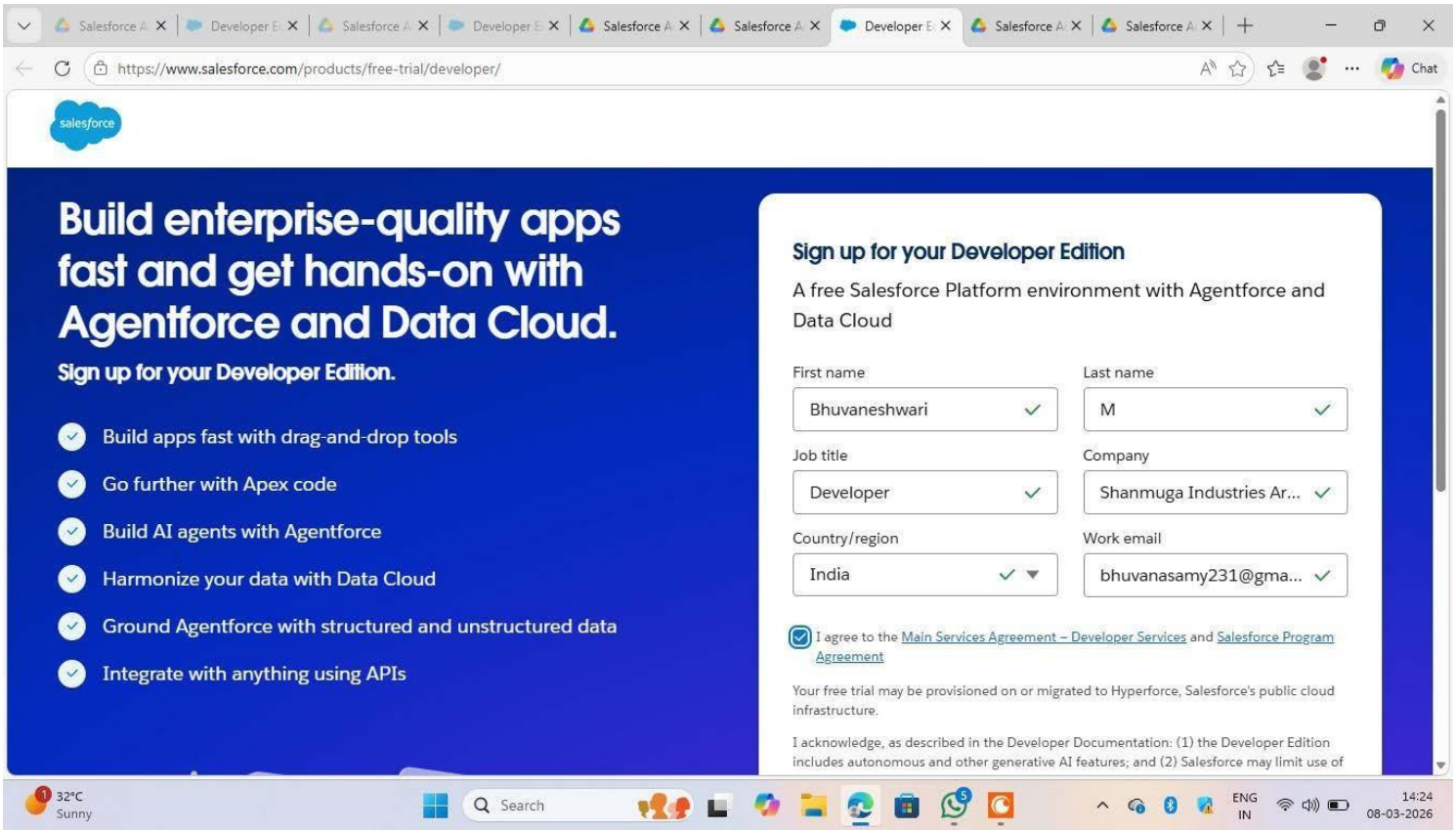
What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers.
Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners,and store your data securely in the cloud.So what does th atreally mean? Well, before Salesforce, your contacts, emails, follow-up tasks, and prospective deals might have been organized something like this:
https://youtu.be/r9EX3lGde5k

Activity1:Creating Developer Account:

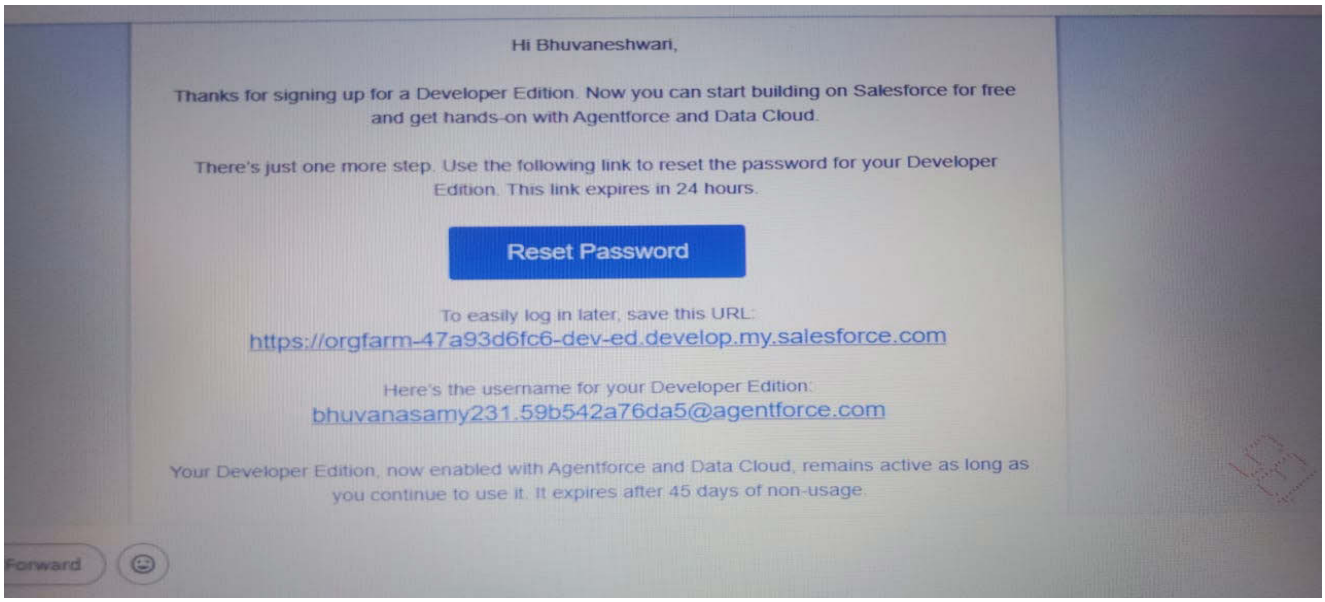
Creating a developer org in salesforce.

Go to https://developer.salesforce.com/signupOnthesignupform,enterthefollowingdetails:

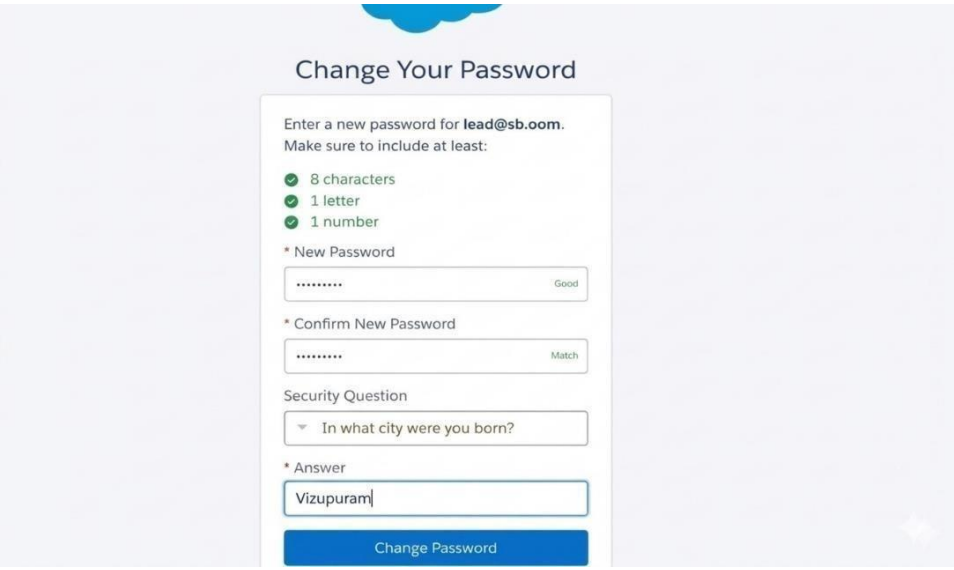


Activity2:Account Activation:

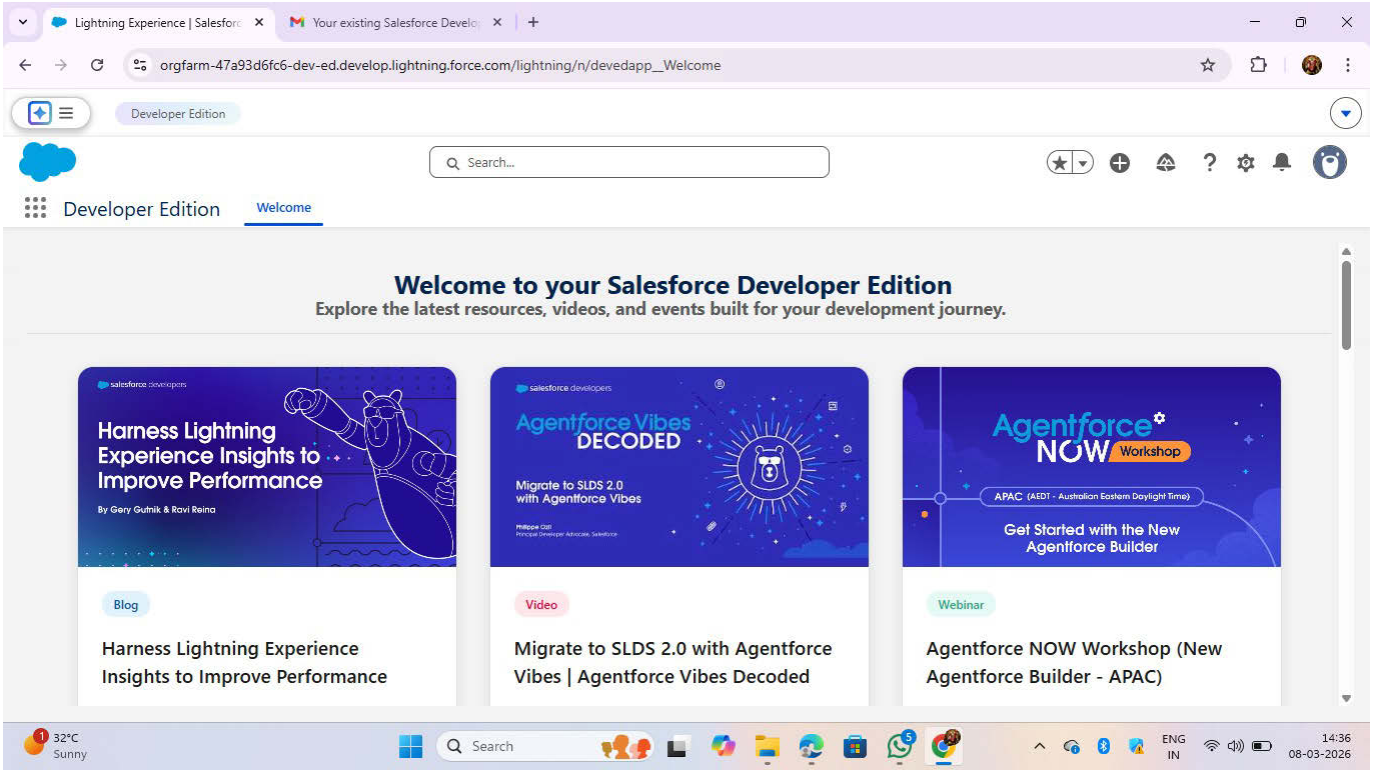
1. Goto the inbox of the email that you used while signingup.Clickonthe verify account to activate your account. The email may take 5-10mins.



2. Clickon VerifyAccount
3. Give a password and answer a security question and click on change password.

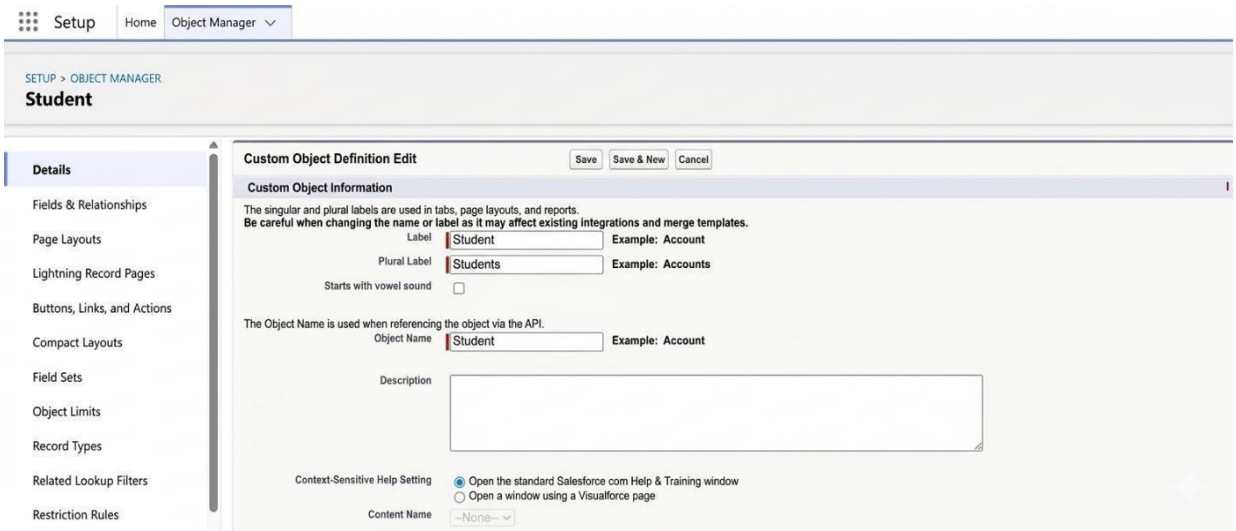


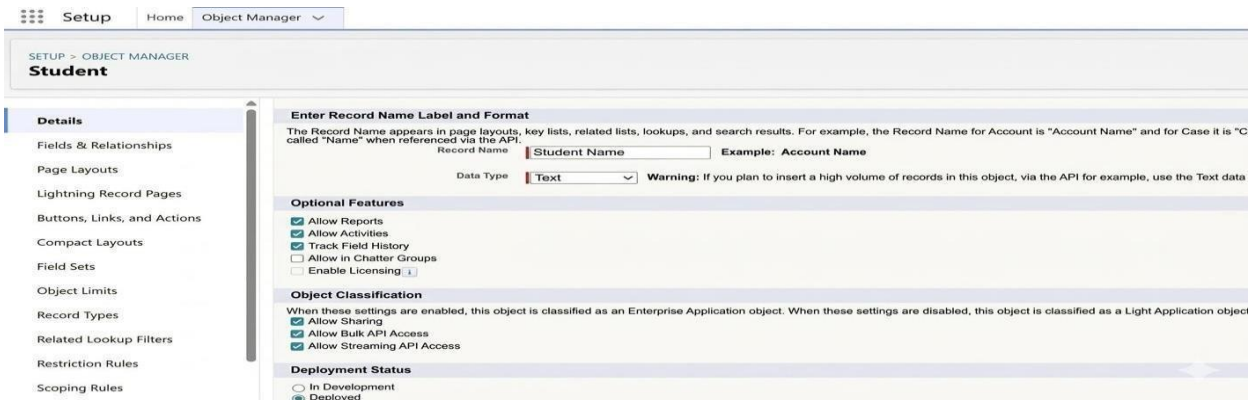
4. Then you will redirect to your salesforce setup page.

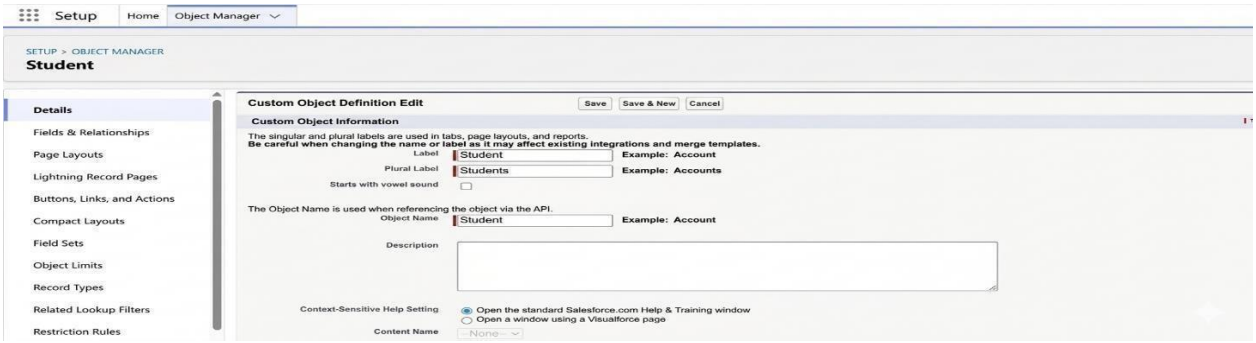


Milestone2 –Objects:

Activity1:Creating Student Object:







In the same way Create Courses,Instructors,**and** Enrolments objects

:

Milestone3 -Fields:

Table1:Student Object Fields

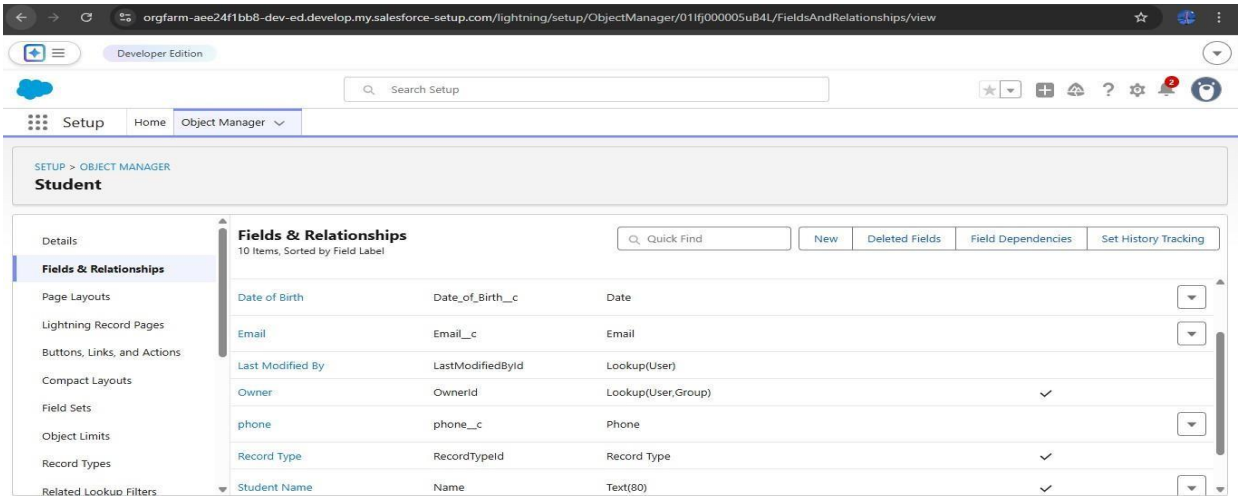


Table 2: course Object Fields

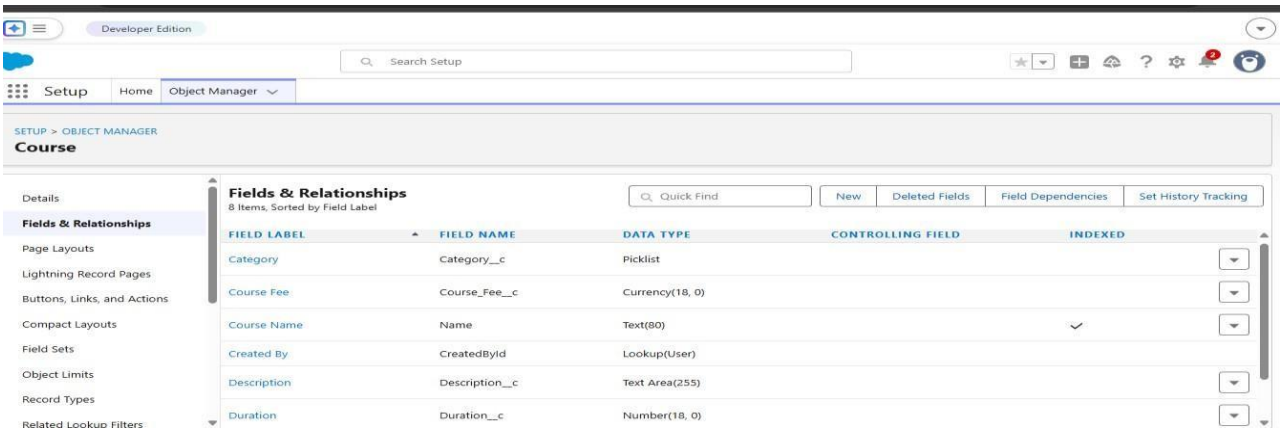
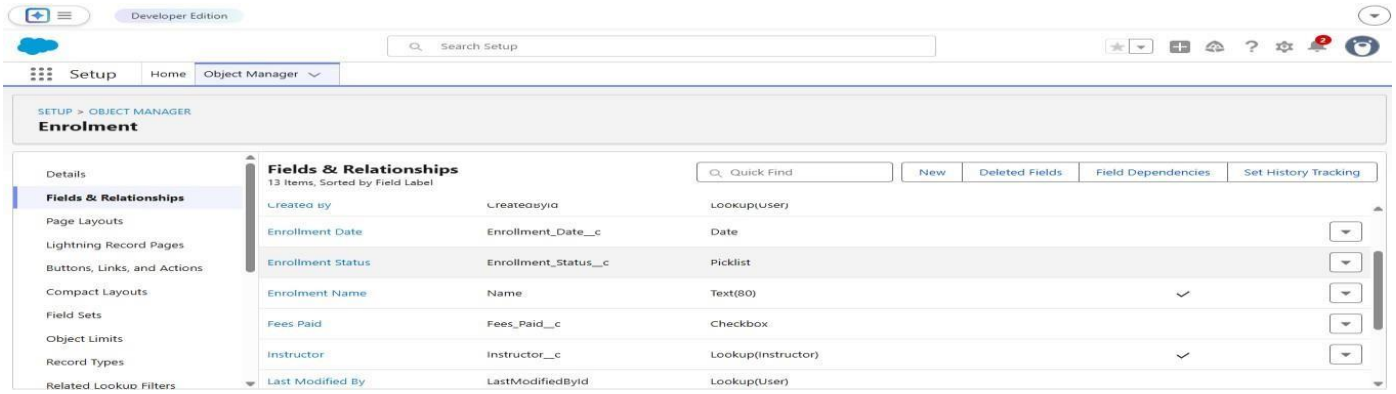


Table 2:Enrolment Object Fields



Milestone4–CreateRelationships:

Activity 1:

Edit Enrollment Custom Field
Student

Help for this Page

Step 1. Choose the field type — Step 1

Specify the type of information that the custom field will contain.

Next Cancel

Data Type

- None Selected — Select one of the data types below.
- Lookup Relationship — Creates a relationship that links this objectto another object. The relationship field allows users to click on a lookup icon to select a value list. The other object is the

Previous Save Next Cancel

Field Label: Student

Field Name: Student

Description:

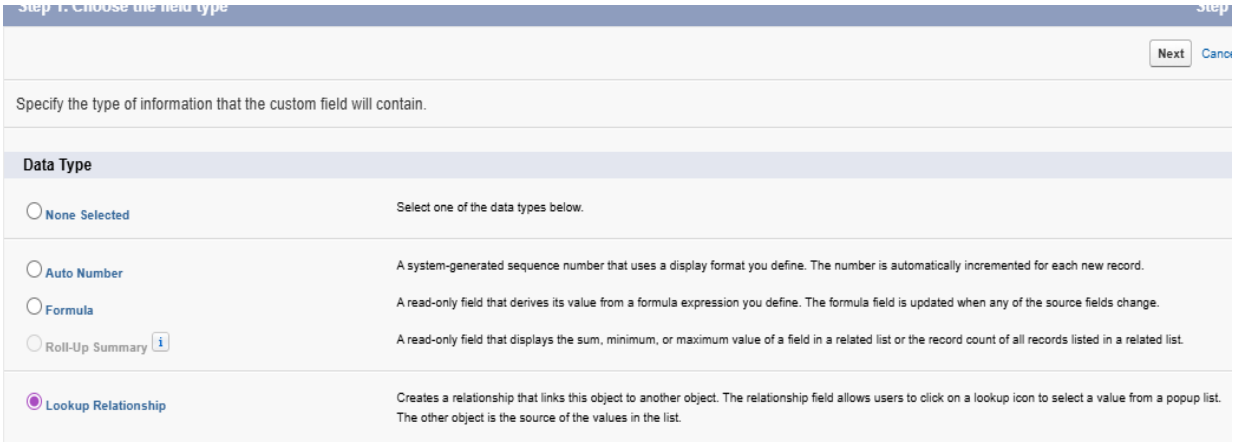
Help Text:

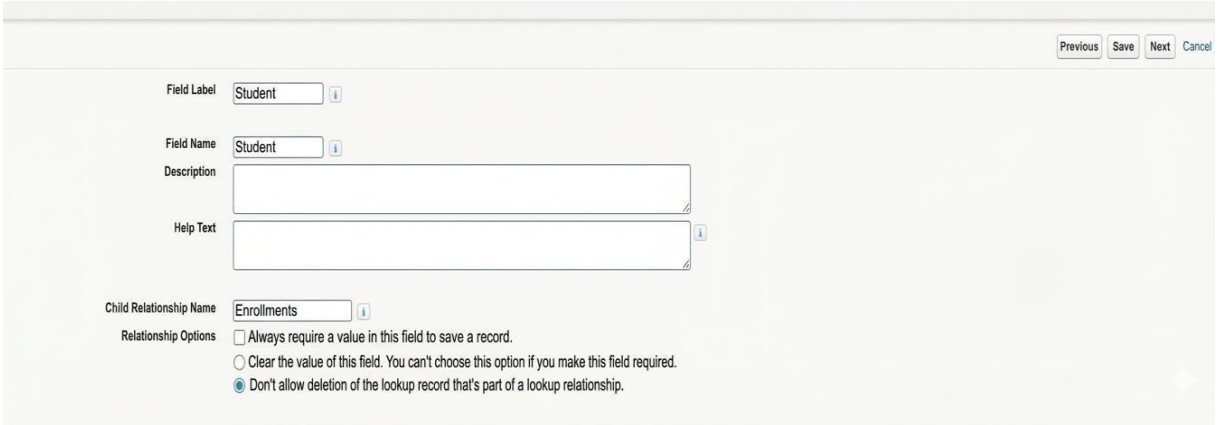
Child Relationship Name: Enrollments

Relationship Options:
- [ ] Always require a value in this field to save a record.
- ( ) Clear the value of this field. You can't choose this option if you make this field required.
- (•) Don't allow deletion of the lookup record that's part of a lookup relationship.

Activity2 :Create Lookup from Enrollment to Course

Purpose:
To link each Enrollment to a Course.

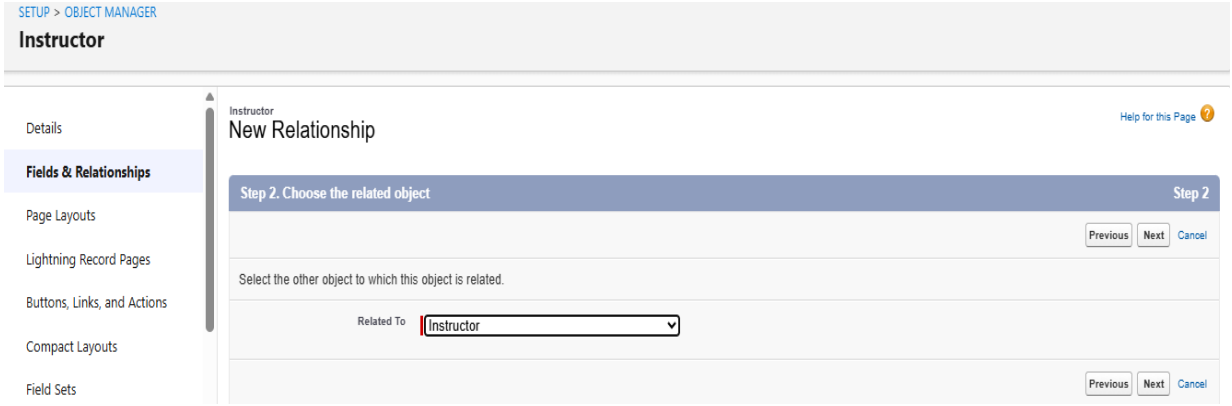


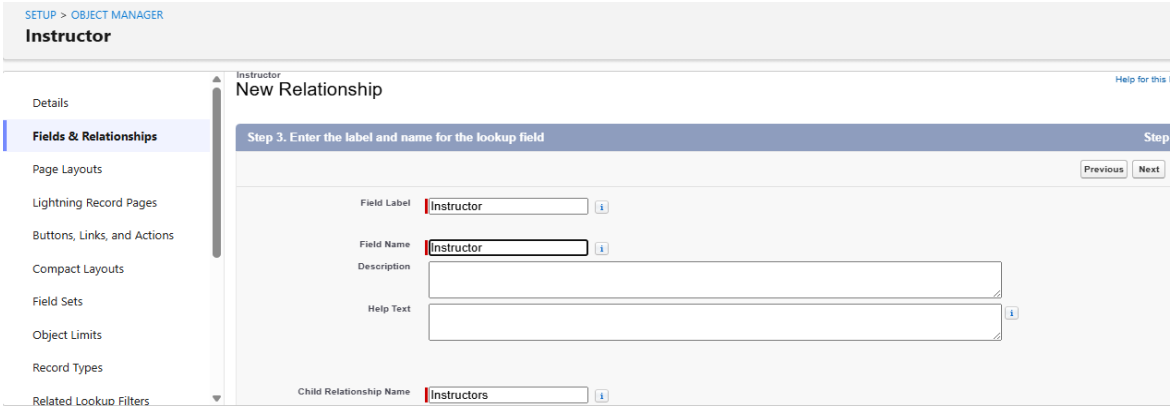


Activity3:Create Lookup from Enrollment to Instructor:

Purpose:

To store which Instructor is assigned to a particular Enrollment.





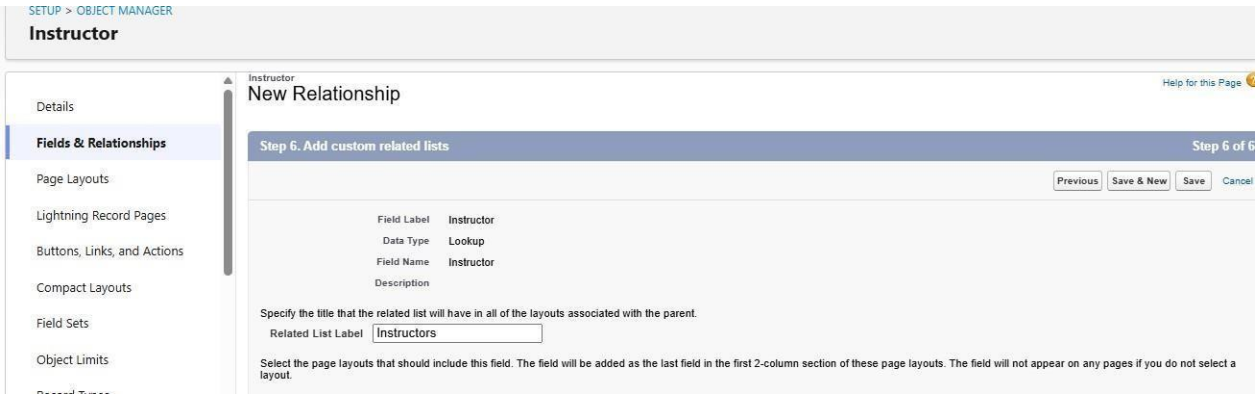
Milestone5–Validation Rules:

Activity1:

Student Email Domain Validation

Purpose:

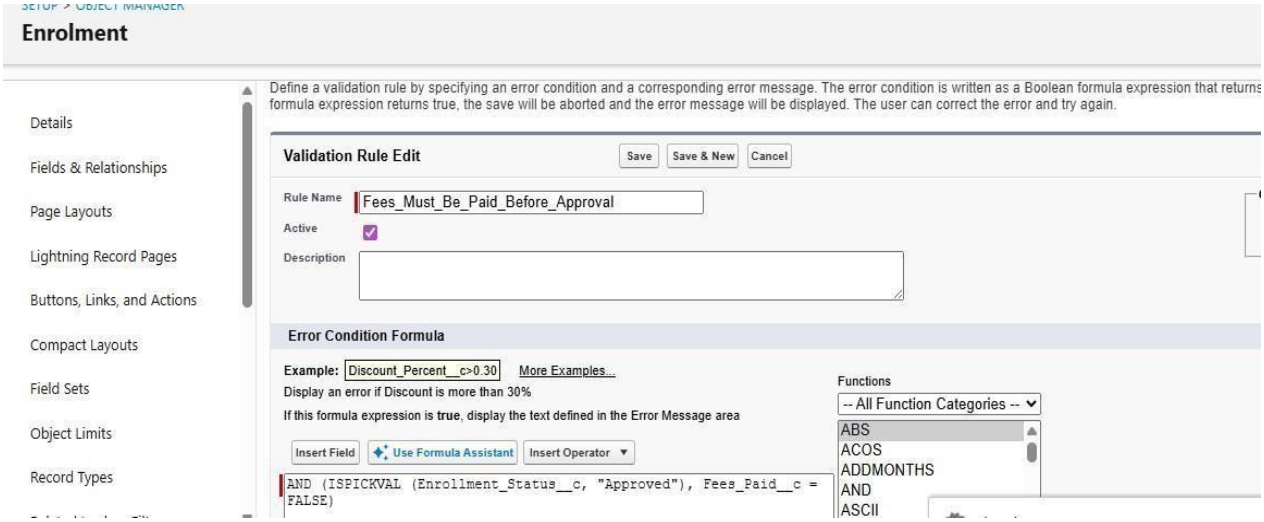
To ensure students use only allowed institutional email



Activity2:Fees Must Be Paid Before Approval

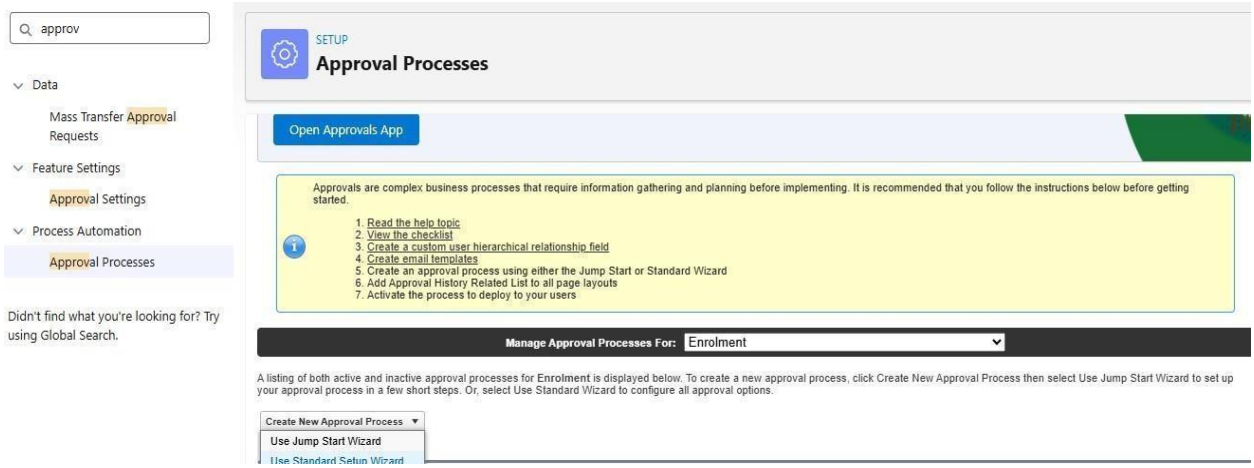
Purpose:

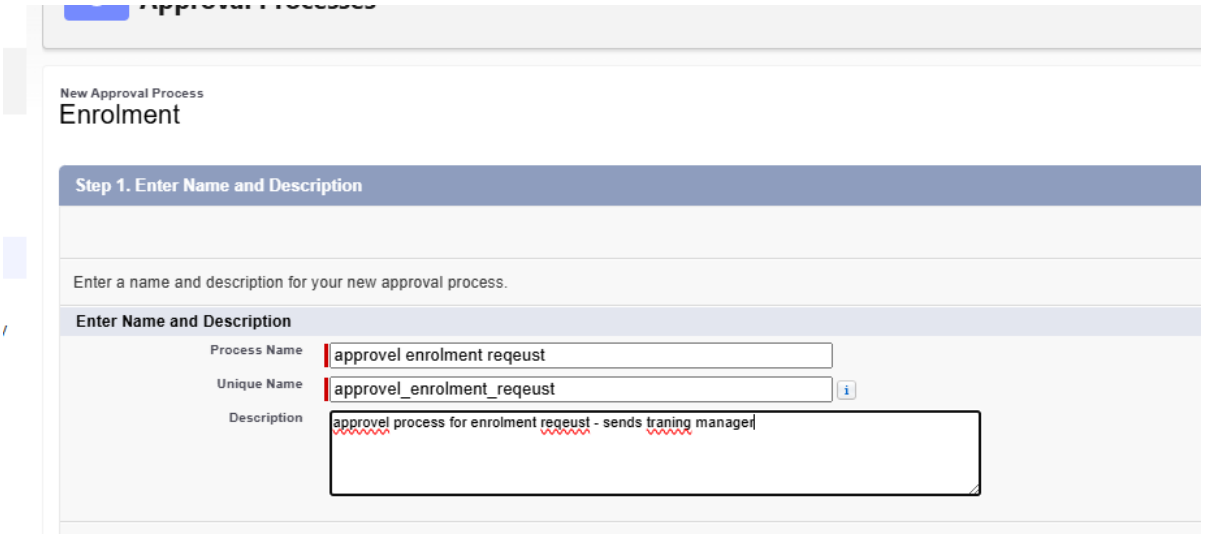
Prevent Enrollment from being approved until fees are paid



Milestone6–Approval Process

Activity1:
Create Approval Processfor Enrollment Requests





Approval Processes

Step 2. Specify Entry Criteria

If only certain types of records should enter this approval process, enter that criteria below. For example, only expense reports from employees at headquarters s
process.

Specify Entry Criteria

Use this approval process if the following criteria are met :

Field	Operator	Value	
Enrolment: Enrollment Status	equals	Requested	AND
--None--	--None--		AND
--None--	--None--		AND
--None--	--None--		AND
--None--	--None--		

Step 3. Specify Approver Field and Record Editability Properties — Step 3 of 6

Previous Next Cancel

When you define approval steps, you can assign approval requests to different users. One of your options is to use a user field to automatically route these requests. If you want to use this option for any of your approval steps, select a field from the picklist below. Also, when a record is in the approval process, it will always be locked-- only an administrator will be able to edit it. However, you may choose to also allow the currently assigned approver to edit the record.

Select Field Used for Automated Approval Routing

Next Automated Approver Determined By: Manager

Use Approver Field of Enrolment Owner: ☐

Record Editability Properties

- (●) Administrators ONLY can edit records during the approval process.
- ( ) Administrators OR the currently assigned approver can edit records during the approval process.

SETUP

Field Updates

Identification

Name: status to approved

Unique Name: status_to_approved

Description:

Object: Enrolment

Field to Update: Enrolment: Enrolment status

Field Data Type: Picklist

Re-evaluate Workflow Rules after Field Change: ☐

Specify New Field Value

Picklist Options

- ( ) The value above the current one
- ( ) The value below the current one
- (●) A specific value: Approved

SETUP

Field Updates

Field Update Edit

Save | Save & New | Cancel

Identification

Name	Status Rejected
Unique Name	Status_Rejected
Description	
Object	Enrolment
Field to Update	Enrolment: Enrolment status
Field Data Type	Picklist
Re-evaluate Workflow Rules after Field Change	☐

Specify New Field Value

Picklist Options

- The value above the current one
- The value below the current one

SETUP

Email Alerts

Create an email alert to associate with one or more workflow rules, approval processes, or entitlement processes. When changing an email alert, any modifications will apply to all rules, approvals, or entitlement processes associated with it.

Email Alert Edit

Save | Save & New | Cancel

Edit Email Alert — * = Required Information

Description	new email alert when contact is created
Unique Name	new_email_alert_when_cont
Object	Contact
Email Template	Welcome Email For Contact
Protected Component	☐
Recipient Type	Search: User for: Find
Recipients	

Available Recipients: User: Integration User, User: OrgFarm EPIC, User: Security User

Selected Recipients: User: Gokulakannan S

Recipients

Available Recipients: User: Integration User, User: OrgFarm EPIC, User: Security User

Add / Remove

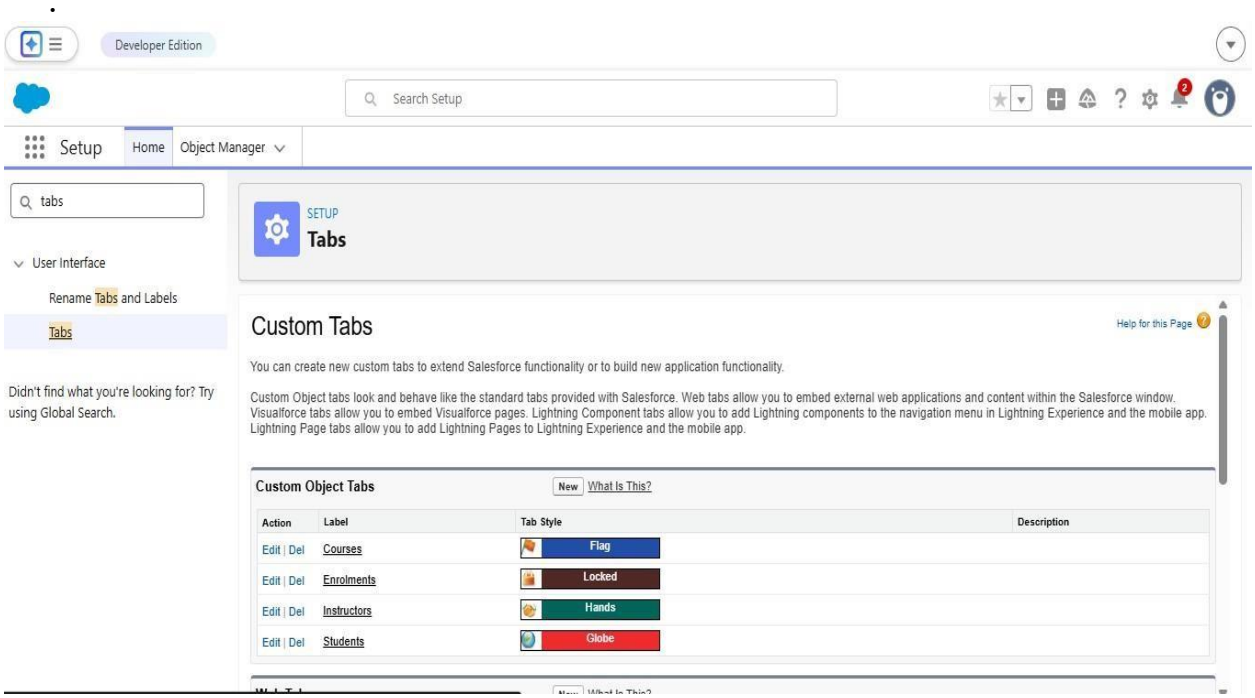
Selected Recipients: User: Gokulakannan S

n enter up to five (5) email addresses to be notified.

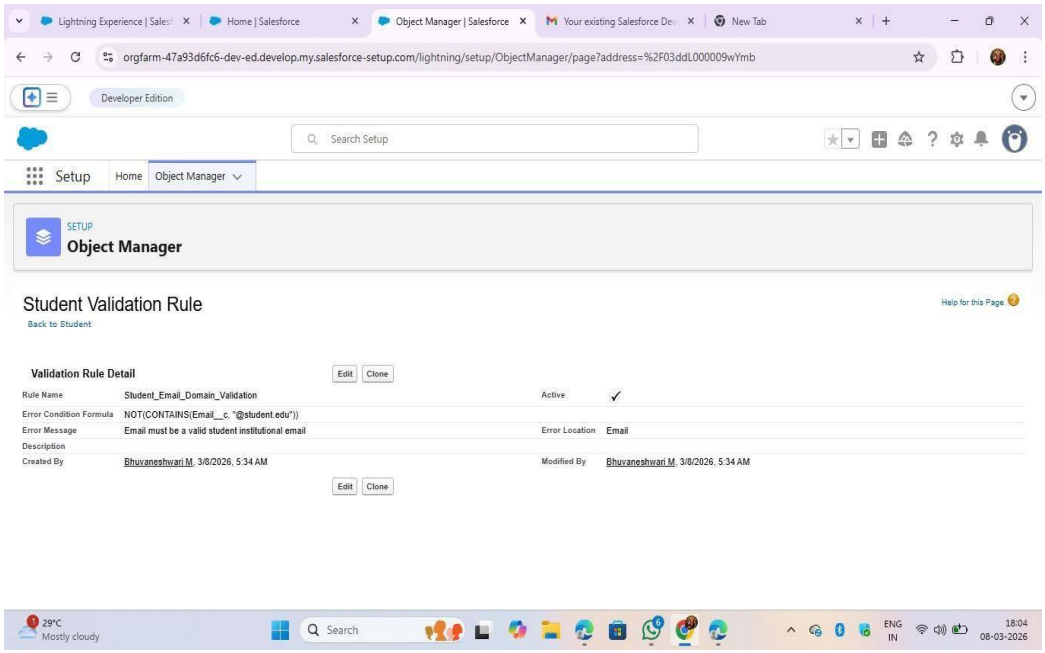
Additional Emails

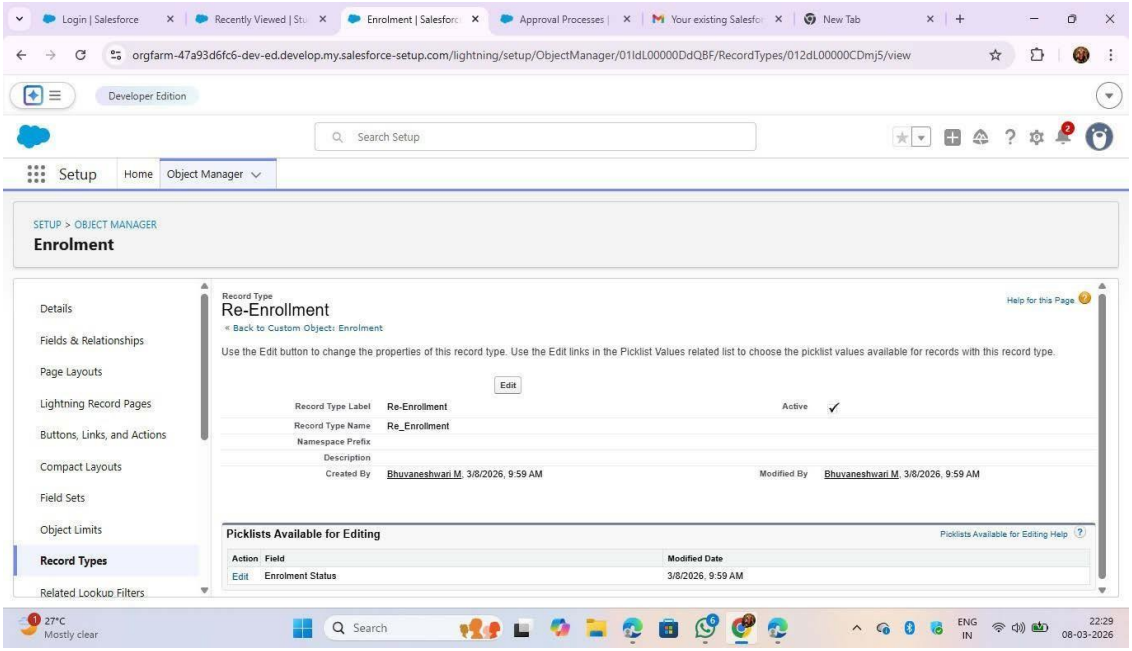
Milestone 7-Tabs

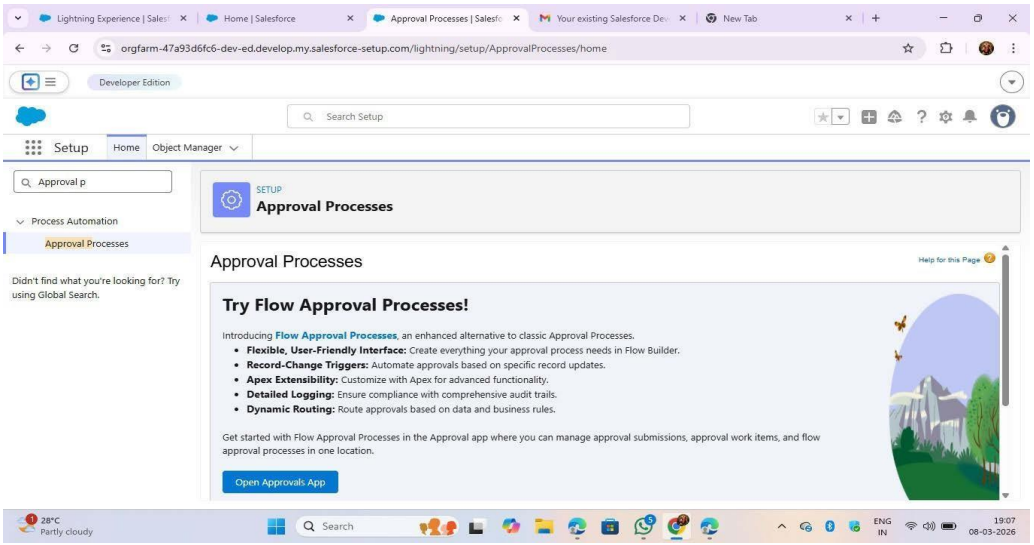
Activity1:Creating aTab for the Student Object:



Activity2:Creating Remaining Tabs




Login | Salesforce
Recently Viewed | Stu
Enrolment | Salesforc
Approval Processes |
Your existing Salesfo
New Tab
orgfarm-47a93d6fc6-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/01IdL00000DdQBF/RecordTypes/012dL00000CDmj5/view
Developer Edition
Search Setup
Setup
Home
Object Manager
SETUP > OBJECT MANAGER
Enrolment
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Record Type
Re-Enrollment
« Back to Custom Objects: Enrolment
Help for this Page
Use the Edit button to change the properties of this record type. Use the Edit links in the Picklist Values related list to choose the picklist values available for records with this record type.
Edit
Record Type Label
Re-Enrollment
Active
Record Type Name
Re_Enrollment
Namespace Prefix
Description
Created By
Bhuvaneshwari M, 3/8/2026, 9:59 AM
Modified By
Bhuvaneshwari M, 3/8/2026, 9:59 AM
Picklists Available for Editing
Picklists Available for Editing Help
Action
Field
Modified Date
Edit
Enrolment Status
3/8/2026, 9:59 AM
27°C
Mostly clear
Search
ENG
IN
22:29
08-03-2026


Lightning Experience | Salesf
Home | Salesforce
Approval Processes | Salesfo
Your existing Salesforce Dev
New Tab
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Try Flow Approval Processes!
Introducing Flow Approval Processes, an enhanced alternative to classic Approval Processes.
Flexible, User-Friendly Interface: Create everything your approval process needs in Flow Builder.
Record-Change Triggers: Automate approvals based on specific record updates.
Apex Extensibility: Customize with Apex for advanced functionality.
Detailed Logging: Ensure compliance with comprehensive audit trails.
Dynamic Routing: Route approvals based on data and business rules.
Get started with Flow Approval Processes in the Approval app where you can manage approval submissions, approval work items, and flow approval processes in one location.
Open Approvals App
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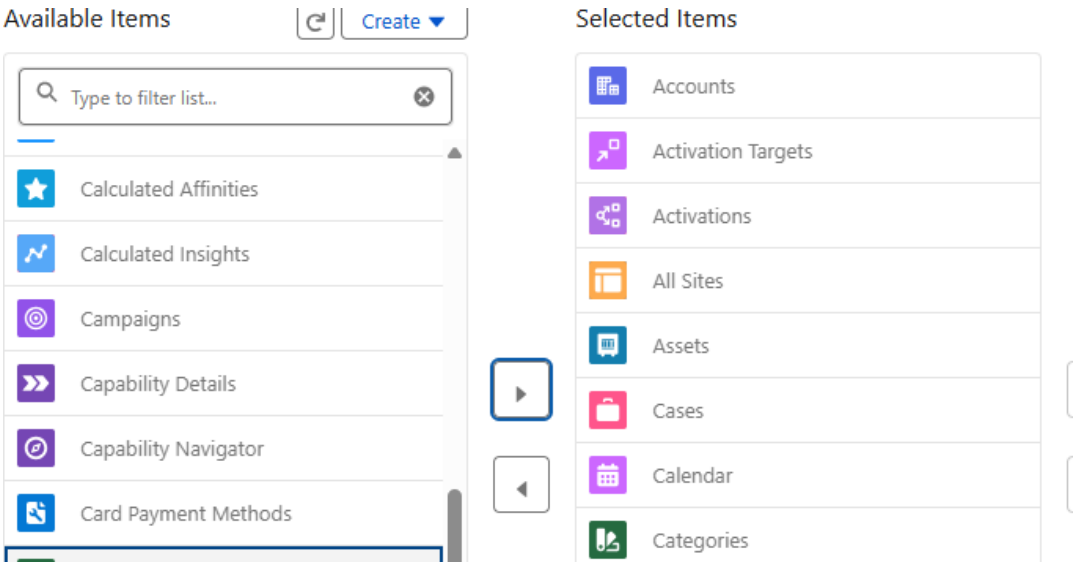
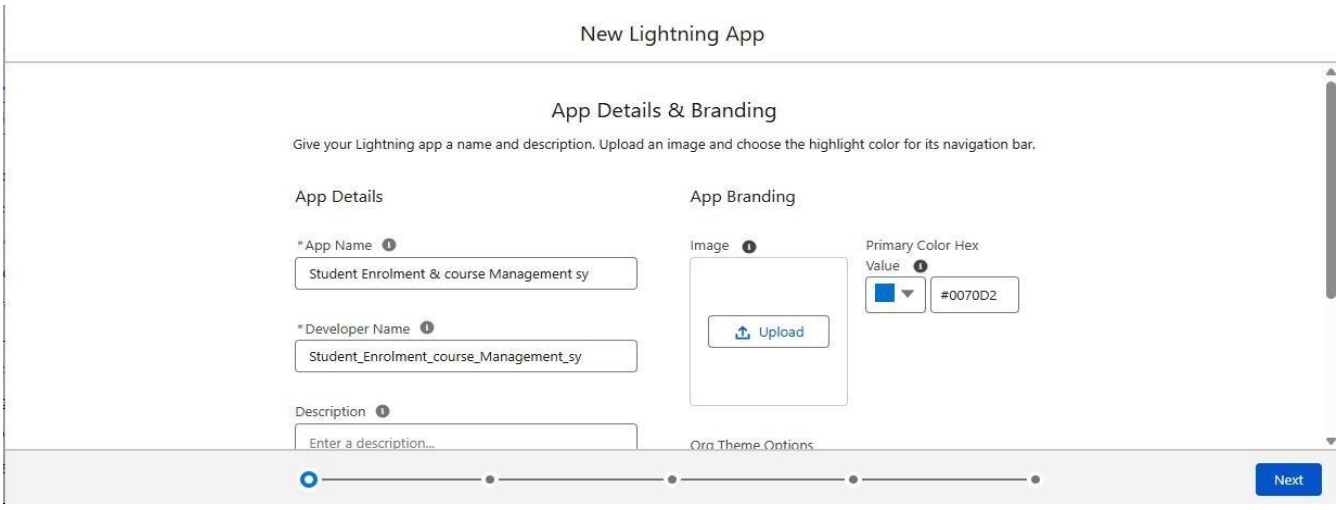
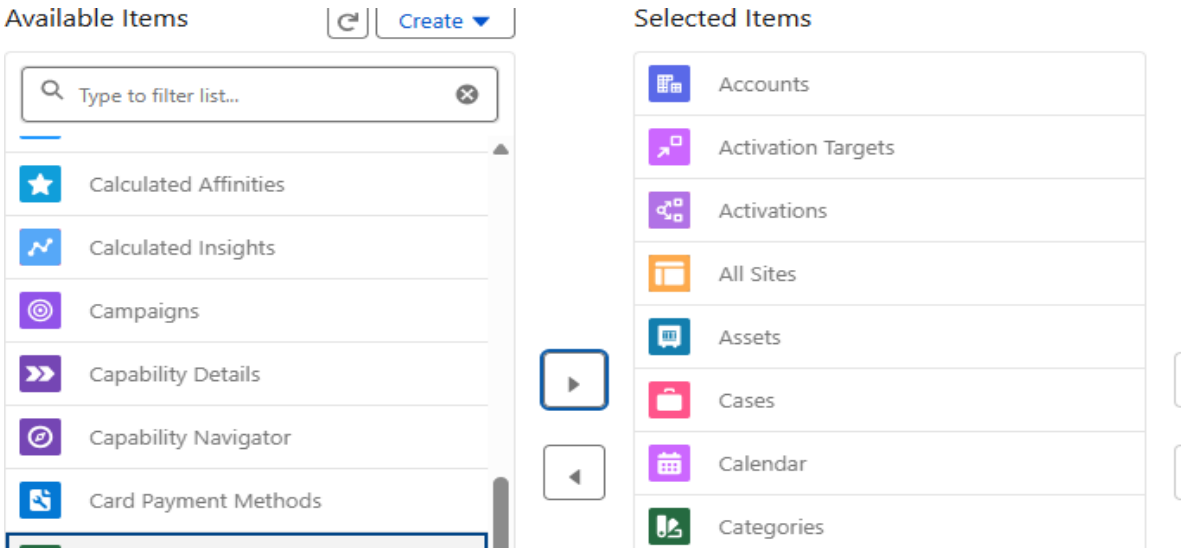
Milestone 8-The Lightning App

Activity1:Creating the Lightning Application for the Project





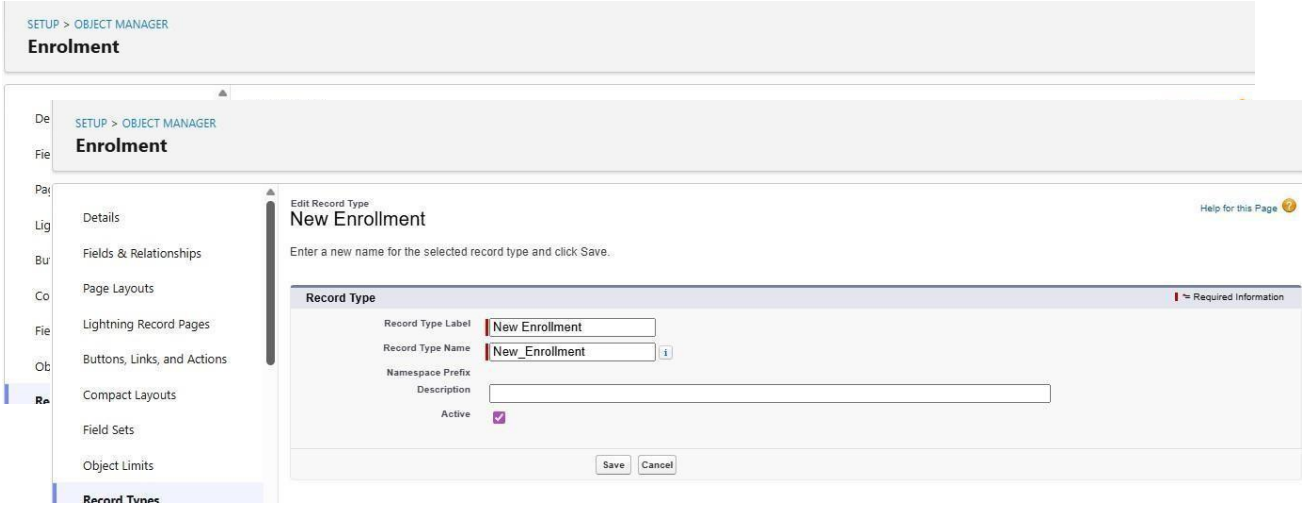

Available Items
Create
Type to filter list...
Calculated Affinities
Calculated Insights
Campaigns
Capability Details
Capability Navigator
Card Payment Methods
Selected Items
Accounts
Activation Targets
Activations
All Sites
Assets
Cases
Calendar
Categories

Milestone 9–Page Layouts and Record:

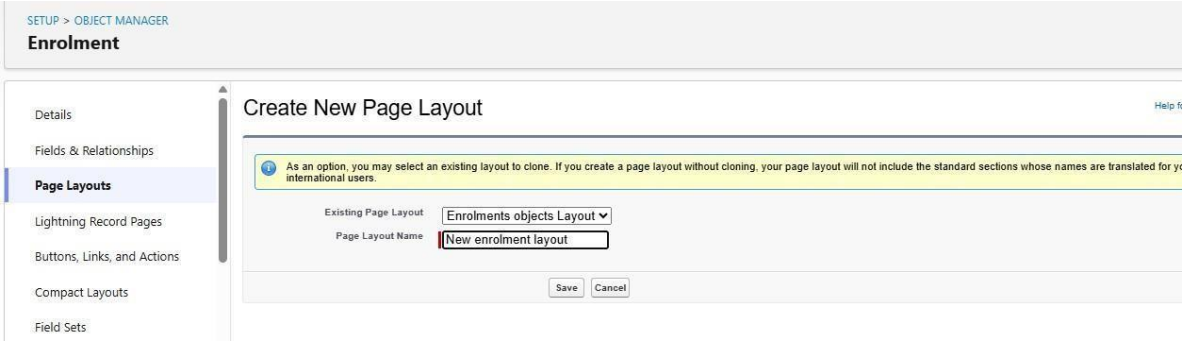
Activity1:Create Record Types for Enrollment:

Purpose:

To separate New Enrollment vs Re-Enrollment processes.



Activity 2: Separate Page Layo Createuts for Each Enrollment Record Type

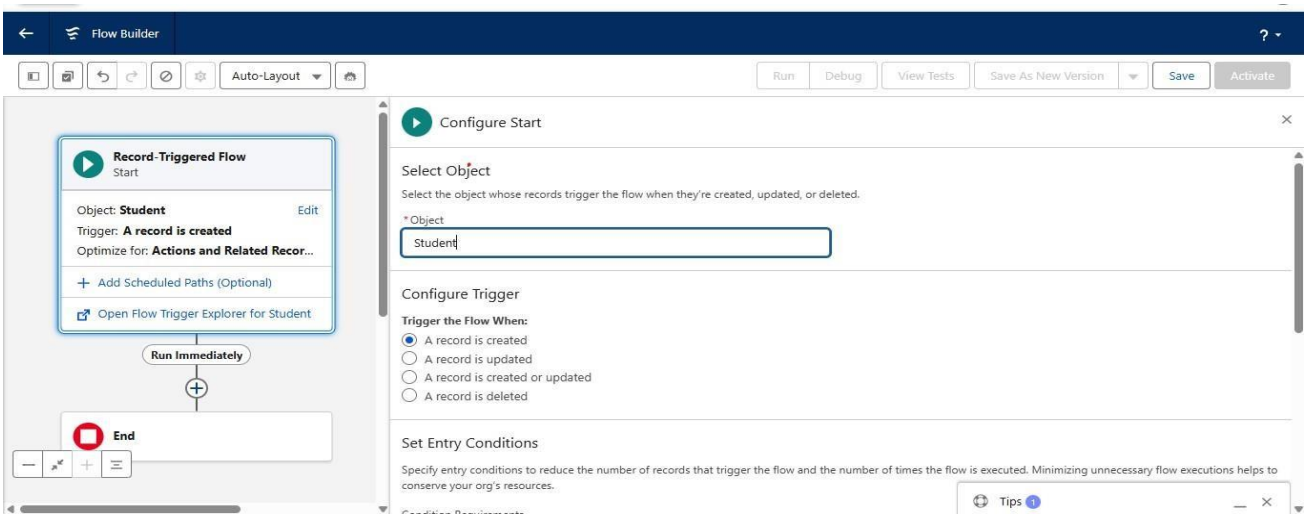


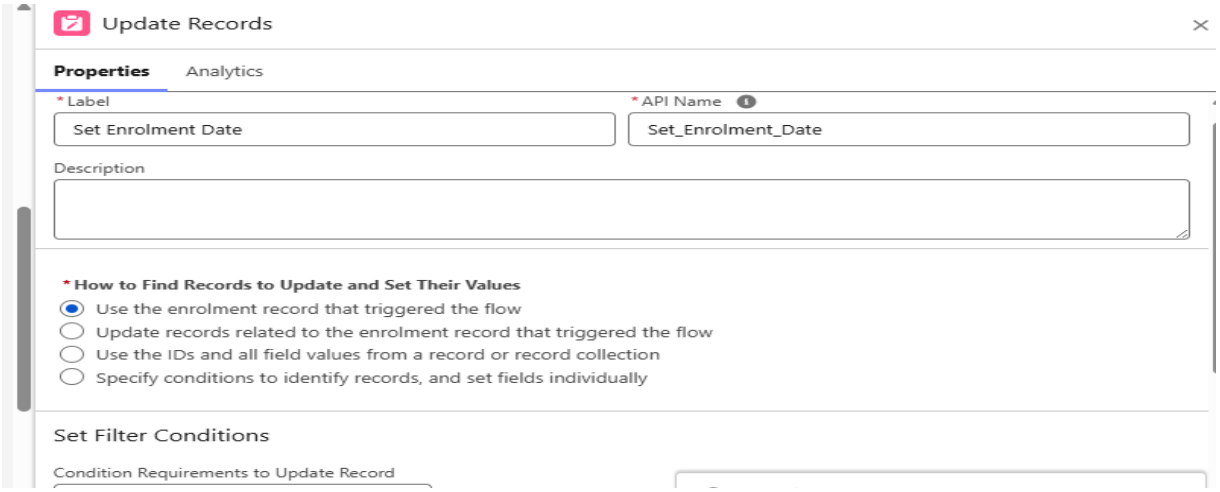
Milestone10–Automation with Flow

Activity1:Auto-Populate Enrollment Date(Record-Triggered Flow)

Purpose:

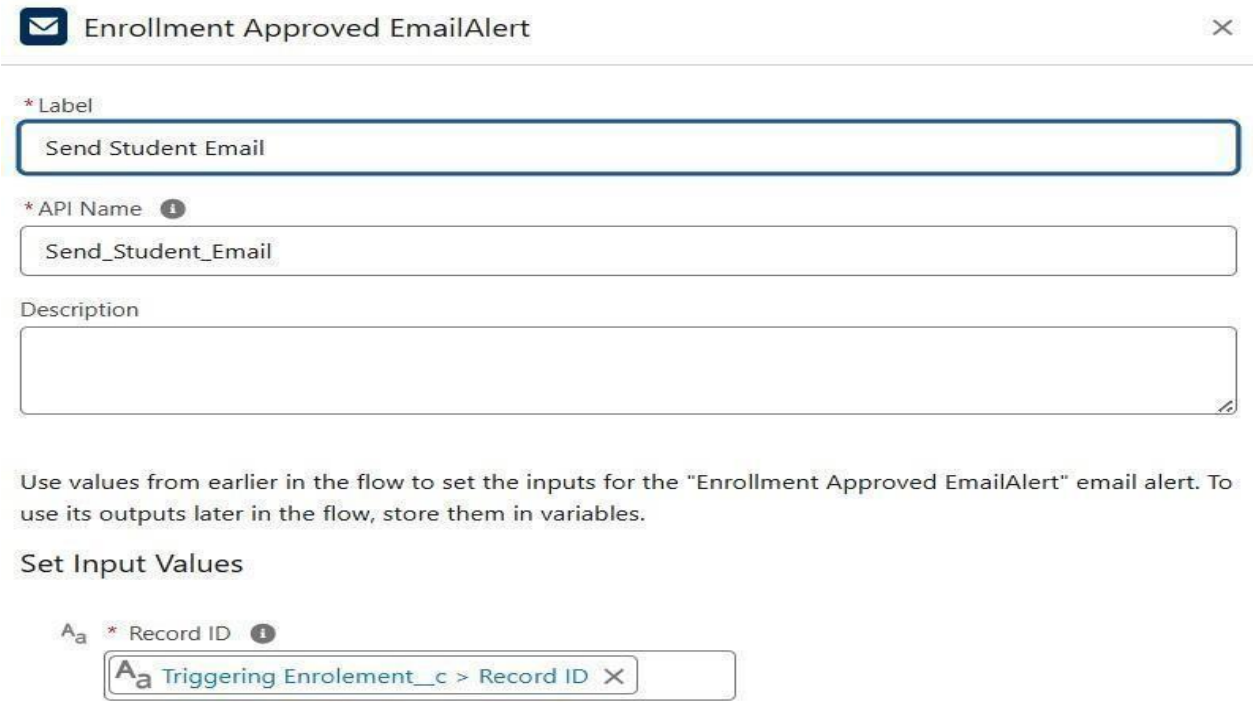
Set Enrollment_Date___c to today when Enrollment is



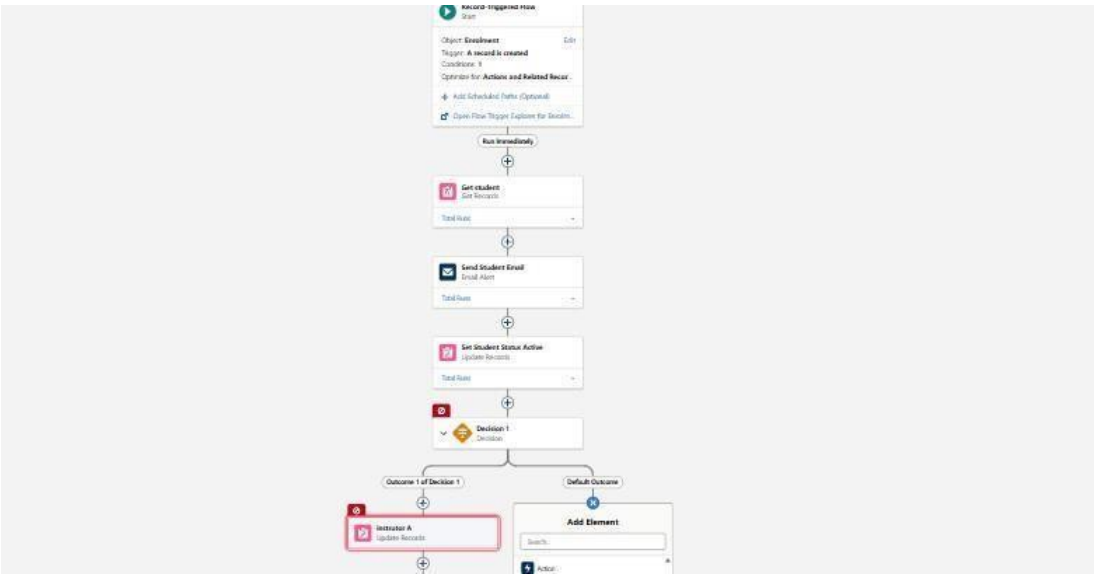


.**Activity 2:** Send Email When Enrollment Approved

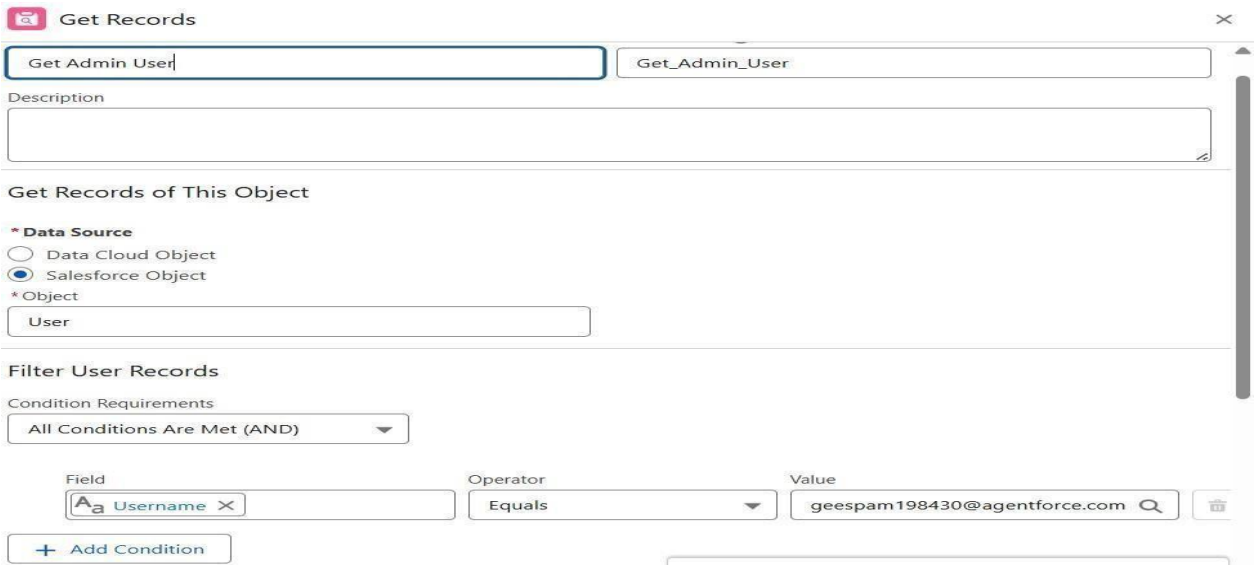
Purpose:
When Enrollment_Status__c=Approved,sendemail,alertadmin,andset Student to Active.

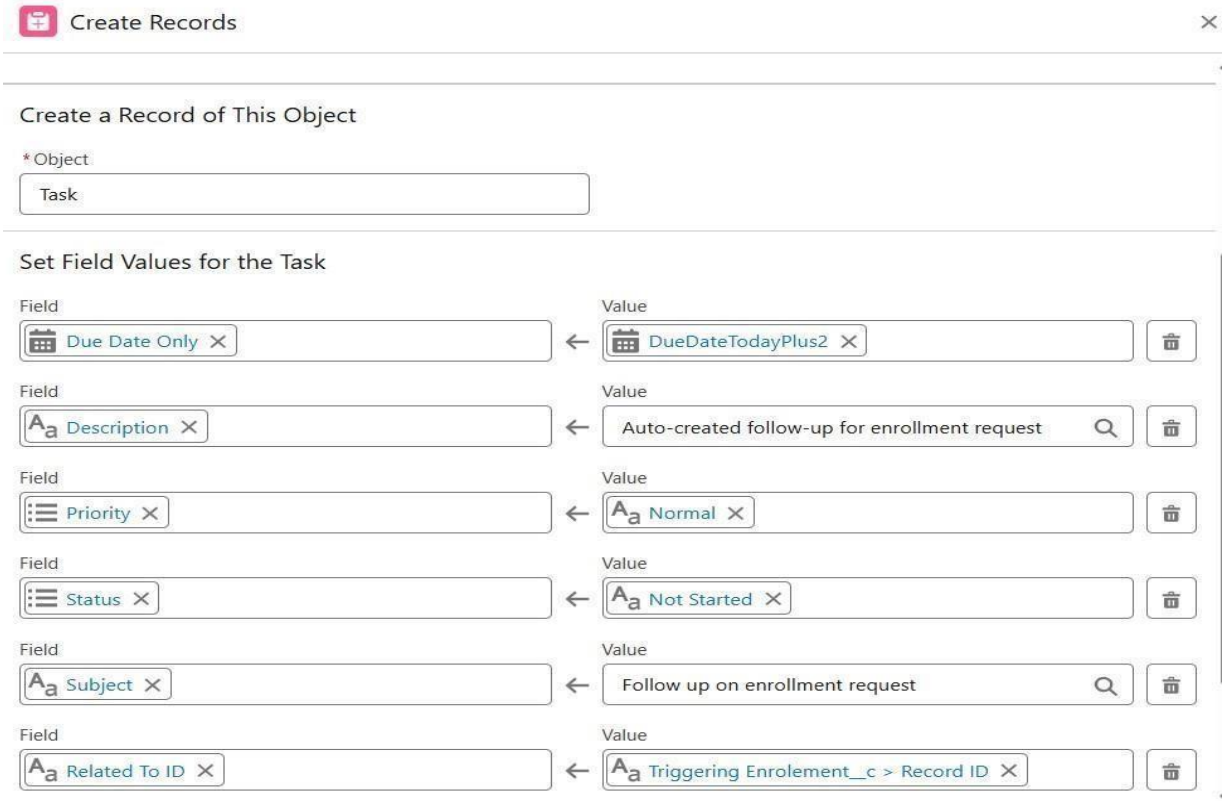


Activity3:Auto-Assign Instructor Based on Course Category



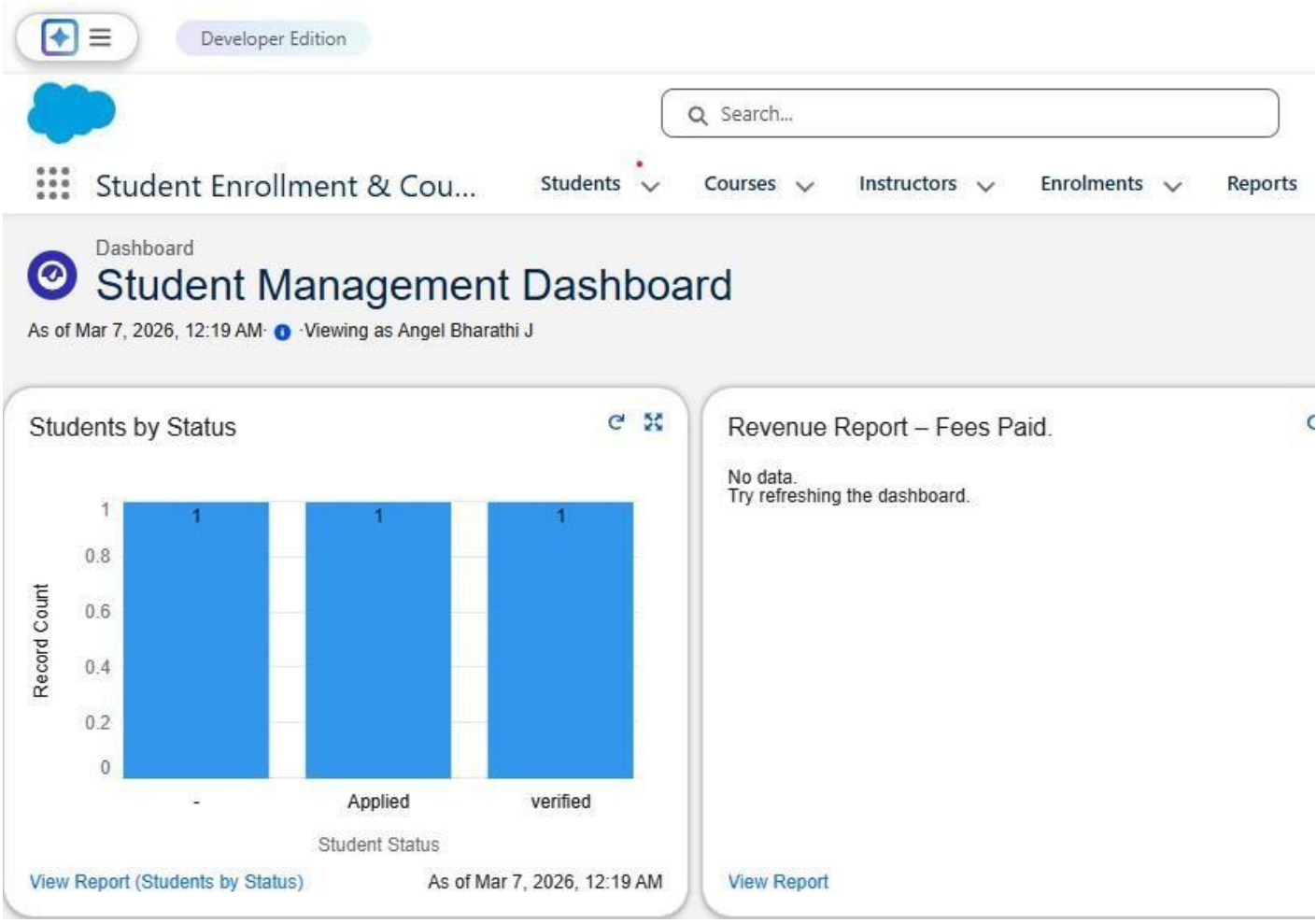
Activity4:Auto-Create Task for Follow-



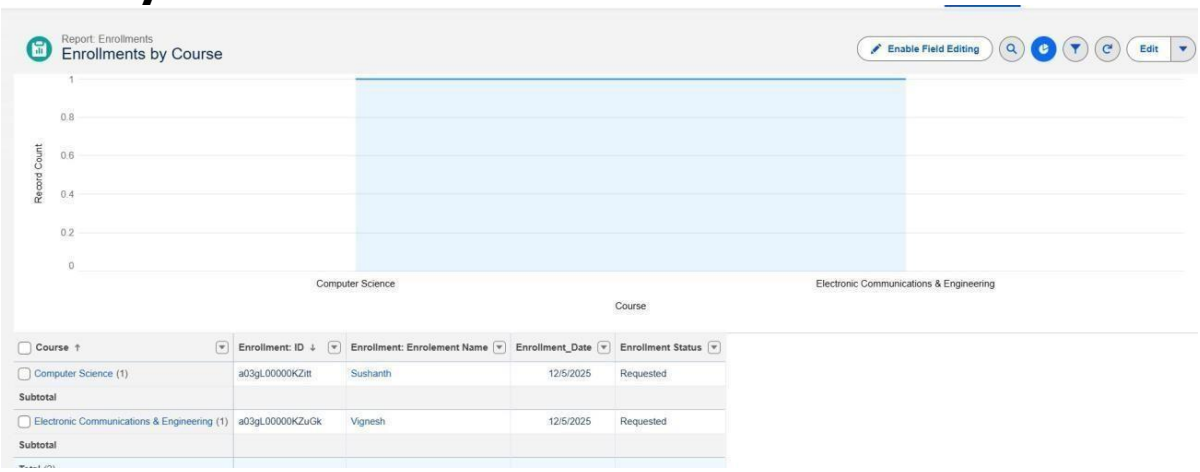


Milestone11–Reports:

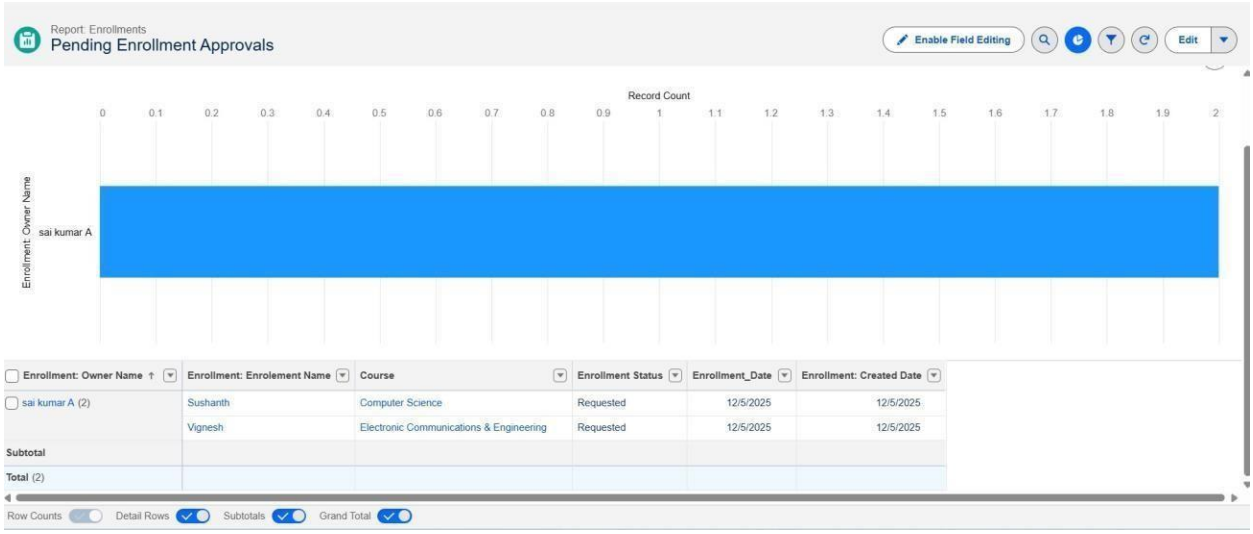
Activity1:Create"Students by Status"Report

1. 

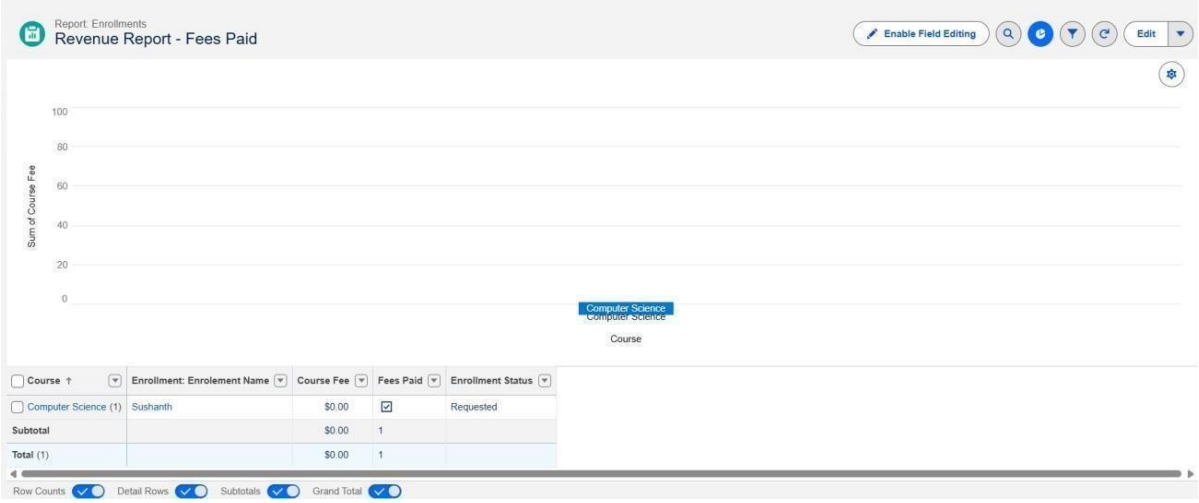
Activity2



Activity3:Create" Pending Enrollment Approvals"Report

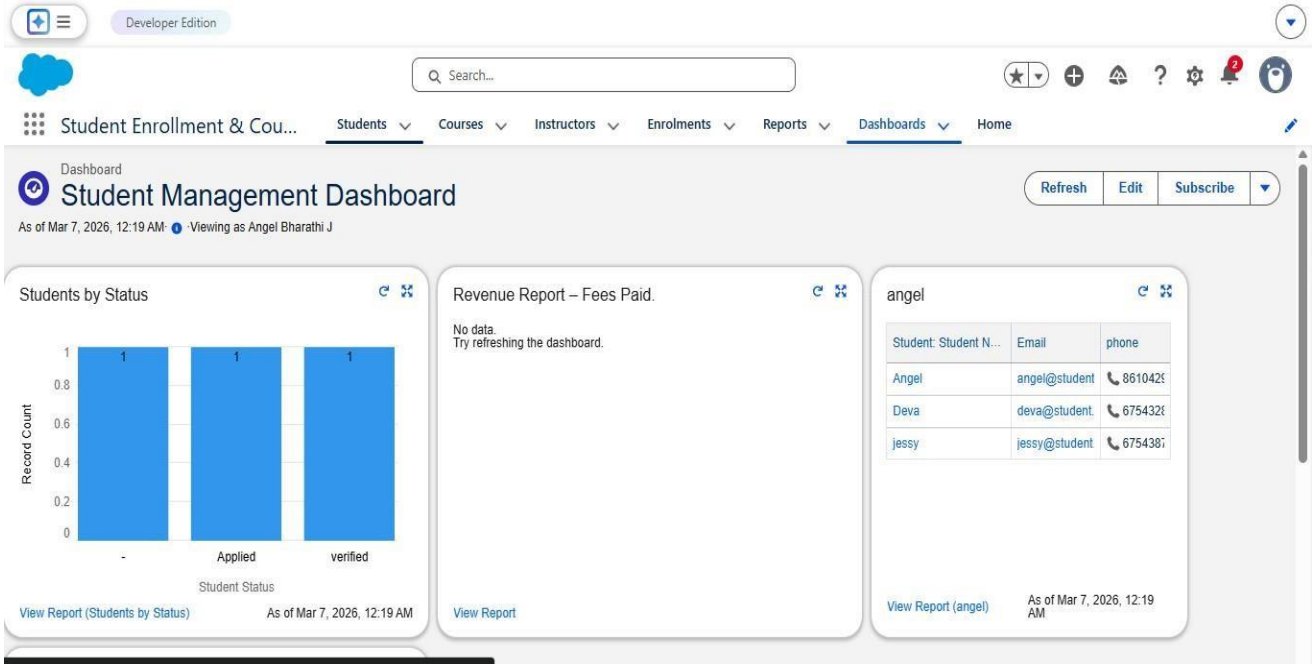


Activity4:Create"Revenue Report(FeesPaid)"Report



Milestone12–Dashboards:

Activity 1:Create"Student Management Dashboard"



Click**Save→Done.**

ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data
- Automated workflows
- Real-time reporting

Disadvantages

- Salesforce dependency
- Requires initial configuration effort

Conclusion:

The Salesforce Student Enrollment & Course Management System successfully streamlines the entire student academic lifecycle—from student registration and course creation to enrollment management, instructor assignment, approvals, revenue tracking, andreporting.

By leveraging Salesforce's automation tools such as validation rules,flows, approval processes, and dashboards, the system improves data accuracy, operational efficiency, and decision-making. This project demonstrates how Salesforce can transform manual academic processes into a scalable, reliable, and user-friendly digital solution.

Key Performance Indicators (KPIs)

- Total student enrollments per semester.
- Course popularity based on enrollment count.
- Revenue generated through course fees.
- Approval processing time for enrollments.
- Number of pending enrollment requests.

Risk Management

- Data Entry Errors – mitigated using validation rules.
- Unauthorized Access – prevented using role-based security and permission sets.
- System Downtime – reduced using Salesforce cloud reliability.
- Data Loss – minimized through automated backups.

Sample Test Cases

- Verify student record creation with valid details.
- Validate rejection of incorrect email domains.
- Confirm that enrollment cannot be approved without fees paid.
- Check automated instructor assignment based on course category.
- Ensure email notification is sent after enrollment approval.

Glossary

- SECMS – Student Enrollment & Course Management System.Enrollment – Process of registering a student in a

course.

- Approval Process – Salesforce automation for record approval.
- Flow – Salesforce automation tool used for business logic.
- KPI – Key Performance Indicator used to measure system success.

Implementation Benefits

- Improved operational efficiency.
- Better student experience.
- Centralized academic data management.
- Faster administrative decision making.
- Scalable solution for growing institution

THANK YOU